COMPASS_Synoptic_LE_Soil-Fluxes_2022_CH4

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0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data

a. i.e. drops fluxes that have the drop column TRUE or FLAG this
b. adjusting times as needed

4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

0.4 Seting up site + dates

0.5 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

```
[1] "CSV file exists and has been read into the code."
```

0.6 Mark fluxes that were retaken - this is the first issue! Need to get all but last flux to be TRUE for drop - fieldnotes with explanation for flux retaken - usually weird zeros, Steven probe not inserted in the soil, or label mistake

##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.7 Match up the metadata to the rest of the data

```
YYYY-MM-DD format failed for start_dates; trying MM/DD/YYYY
```

```
4 entries had no timestamp matches!
```

```
YYYY-MM-DD format failed for start_dates; trying MM/DD/YYYY
```

0.8 fixing some Deadbands Revisit here, it is not working to fix the dead band

```
Warning in left_join(., time_overrides, by = "Plot_ID"): Detected an unexpected many-
to-many relationship between `x` and `y`.
i Row 13802 of `x` matches multiple rows in `y`.
i Row 10 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

##Extracting initial CH4 concentration

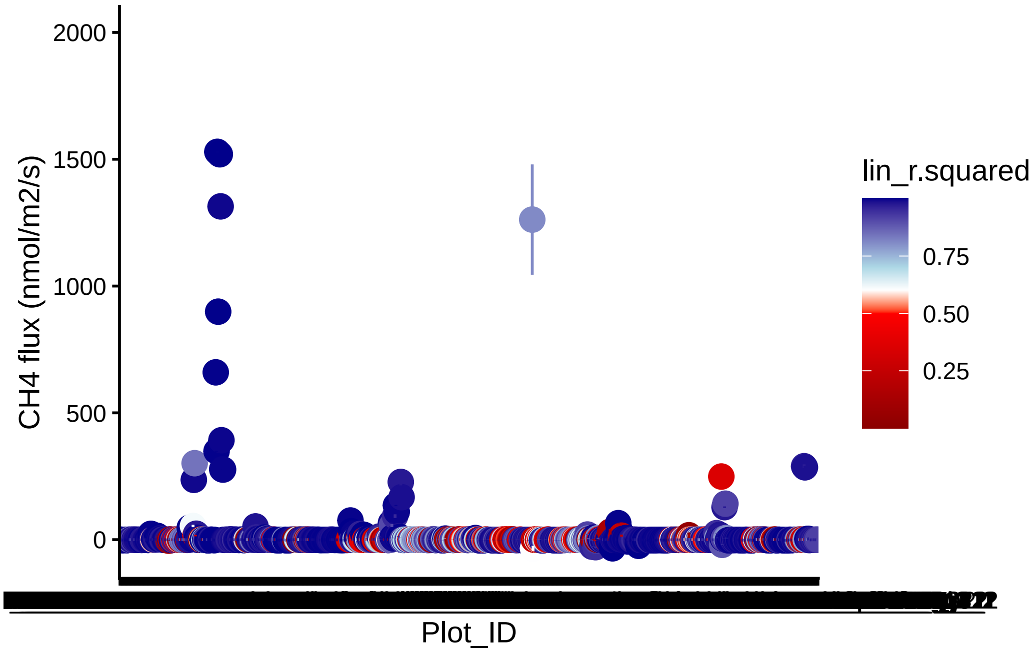
0.9 Change the units from ppb to nmol

```
Assuming atm = 101325 Pa
```

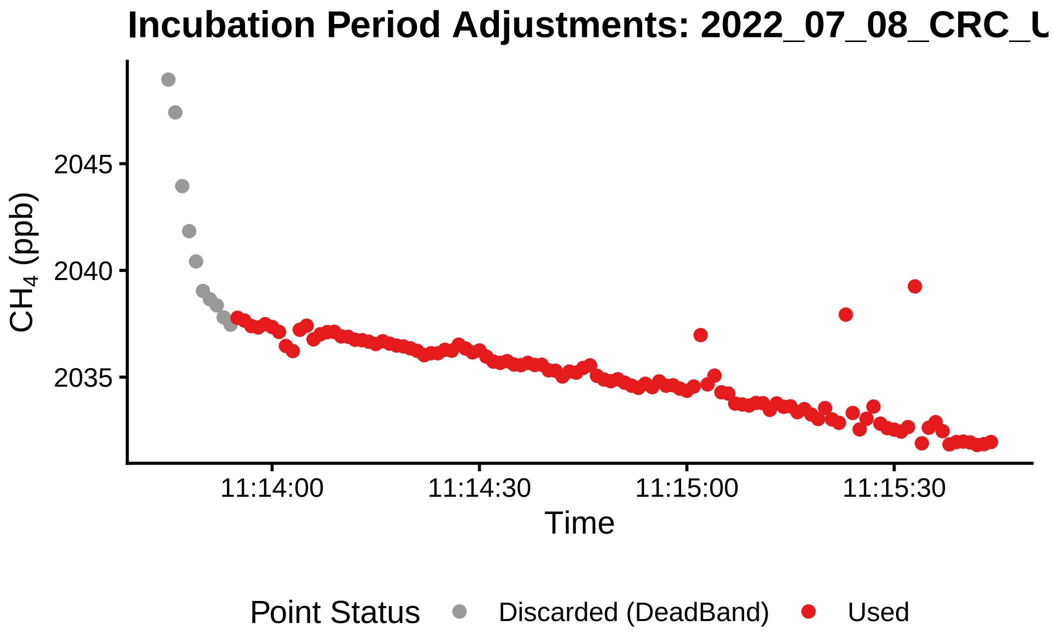
```
Using R = 8.31446261815324 m3 Pa K-1 mol-1
```

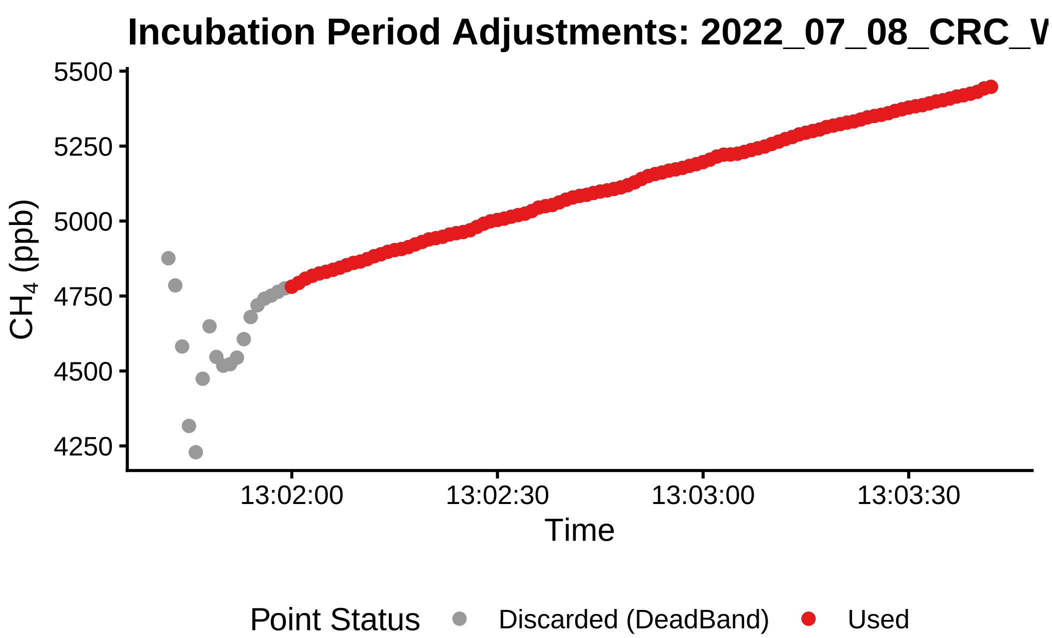
0.10 Calculate fluxes

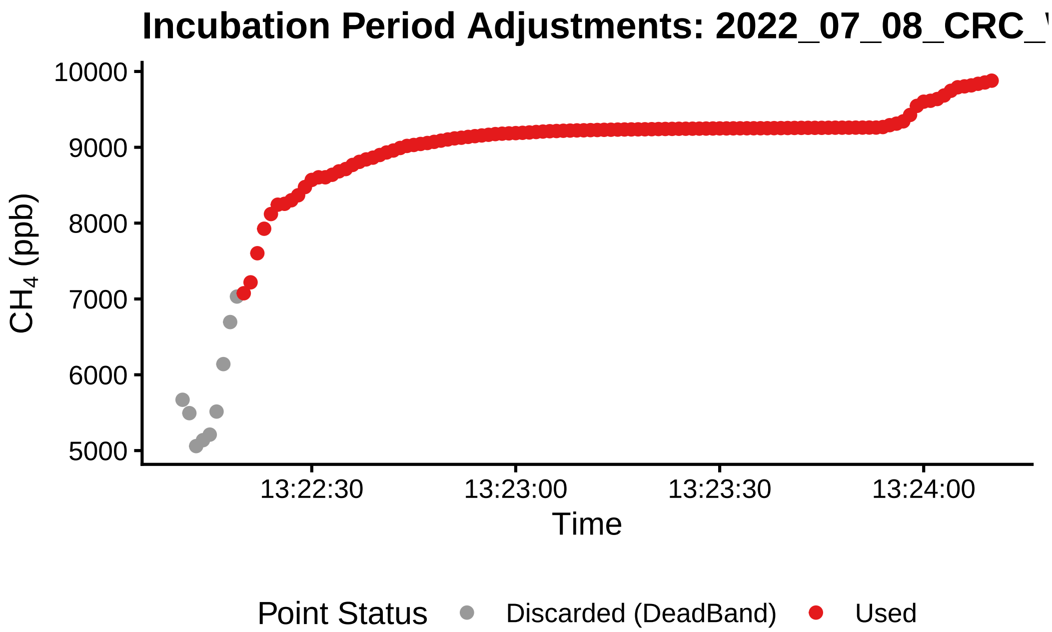
0.11 Visualize fluxes and sort by r2



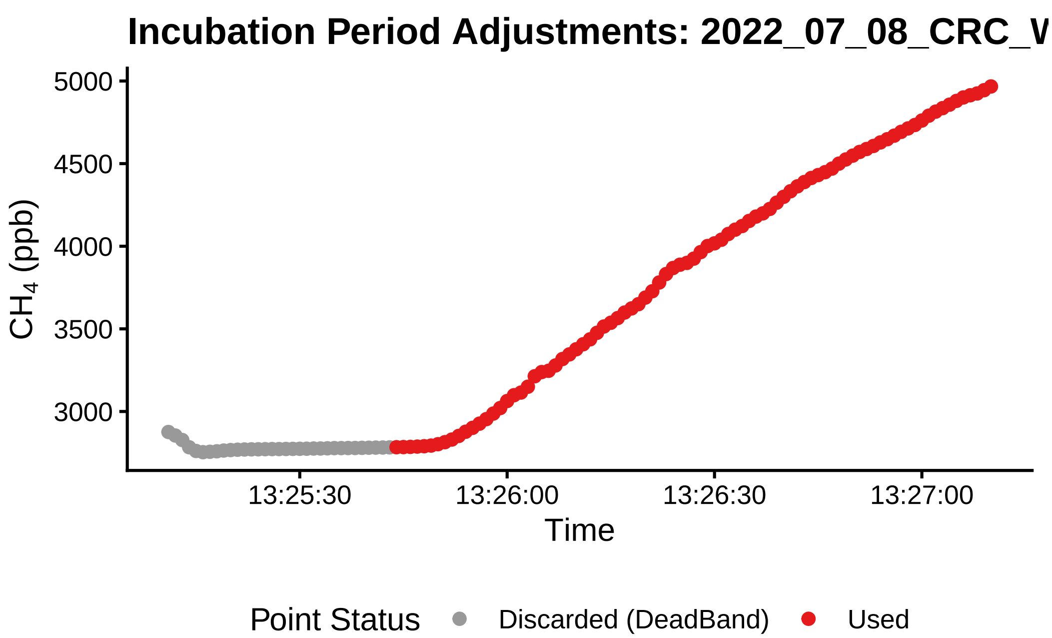
0.12 check the fluxes for which I changed the start time due to lags or high values (recorded in notes of metadata)



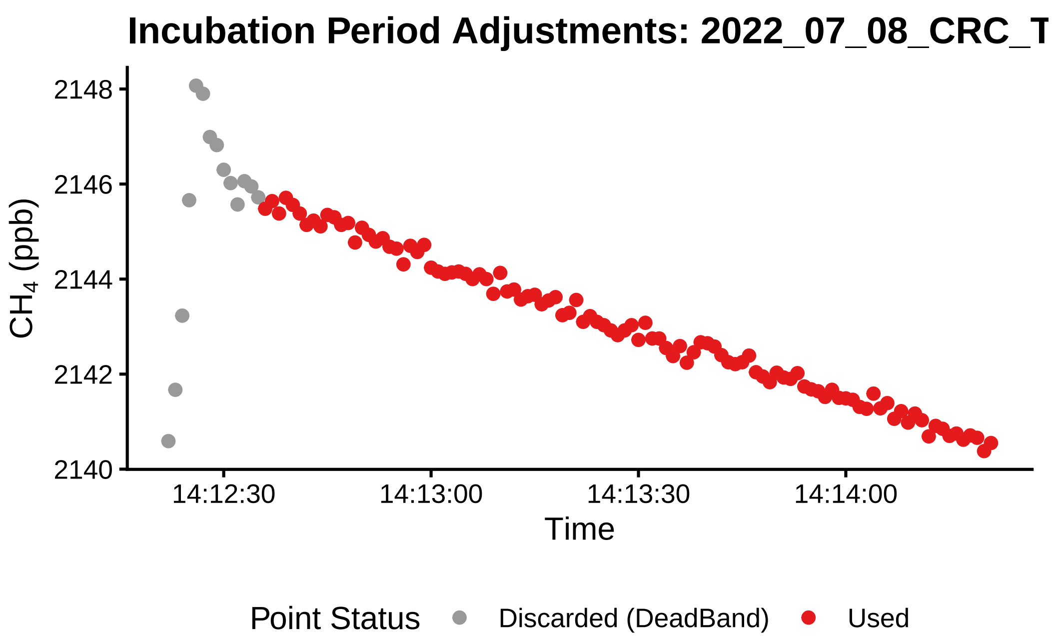


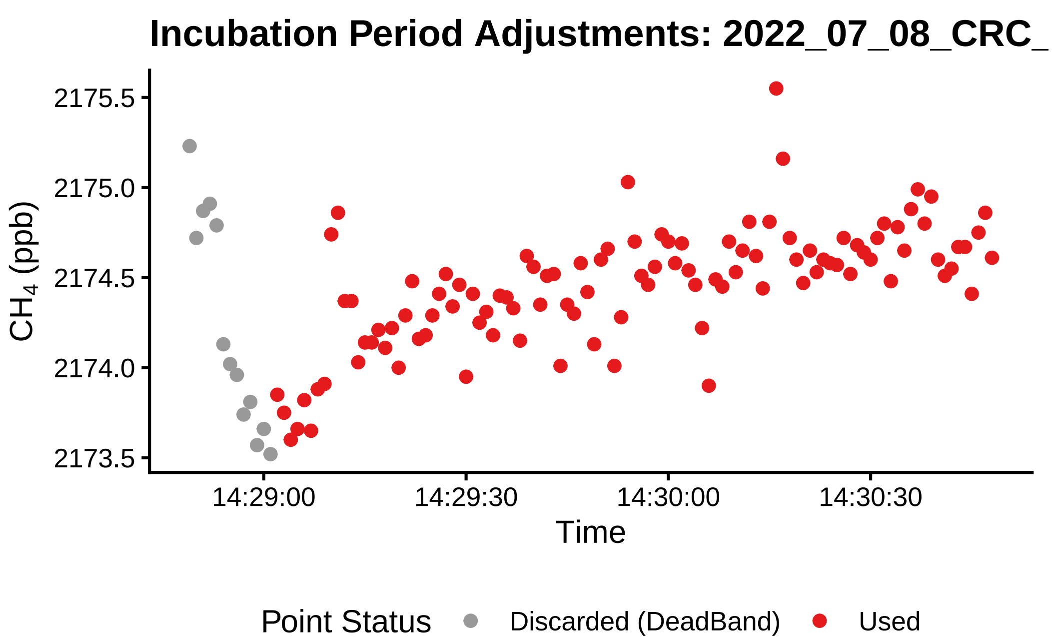


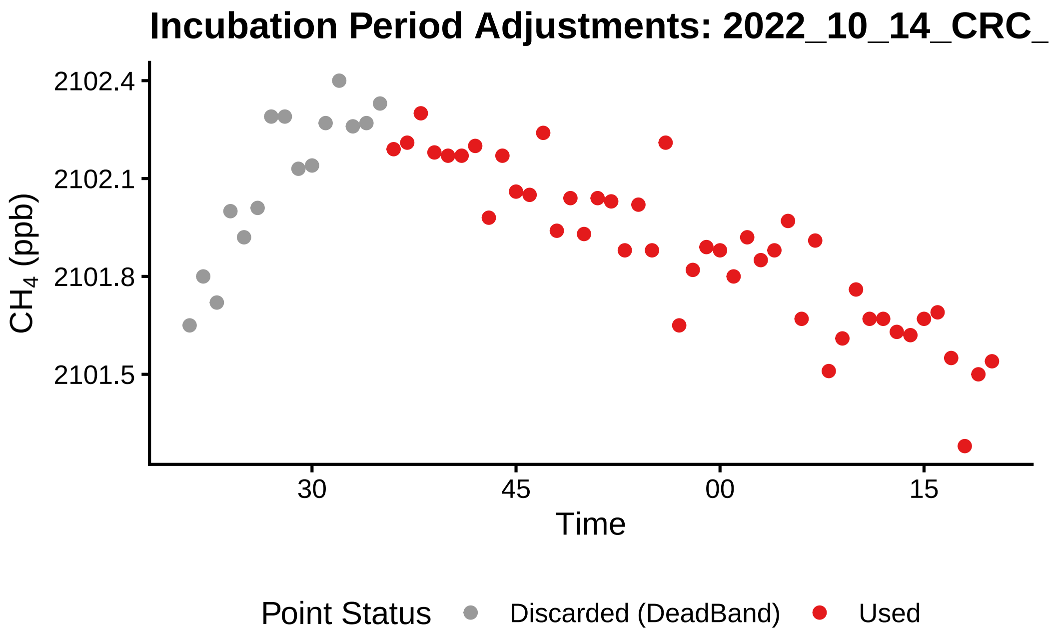
Incubation Period Adjustments: 2022_07_08_CRC_\
10000
9000
8000
7000
6000
5000
CH4 (ppb)
13:22:30
13:23:00
13:23:30
13:24:00
Time
Point Status
Discarded (DeadBand)
Used

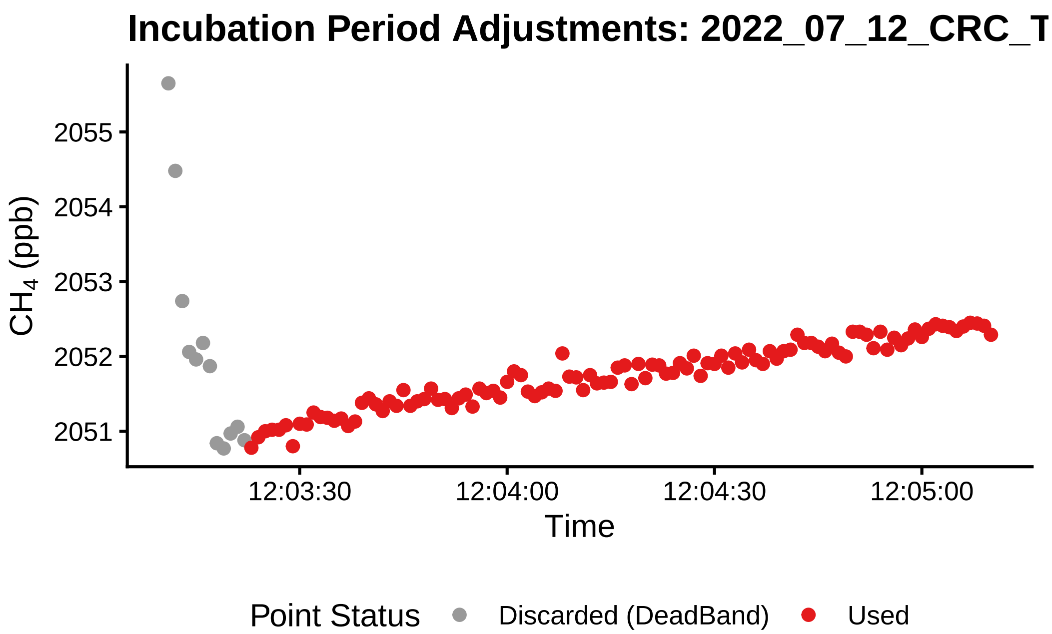


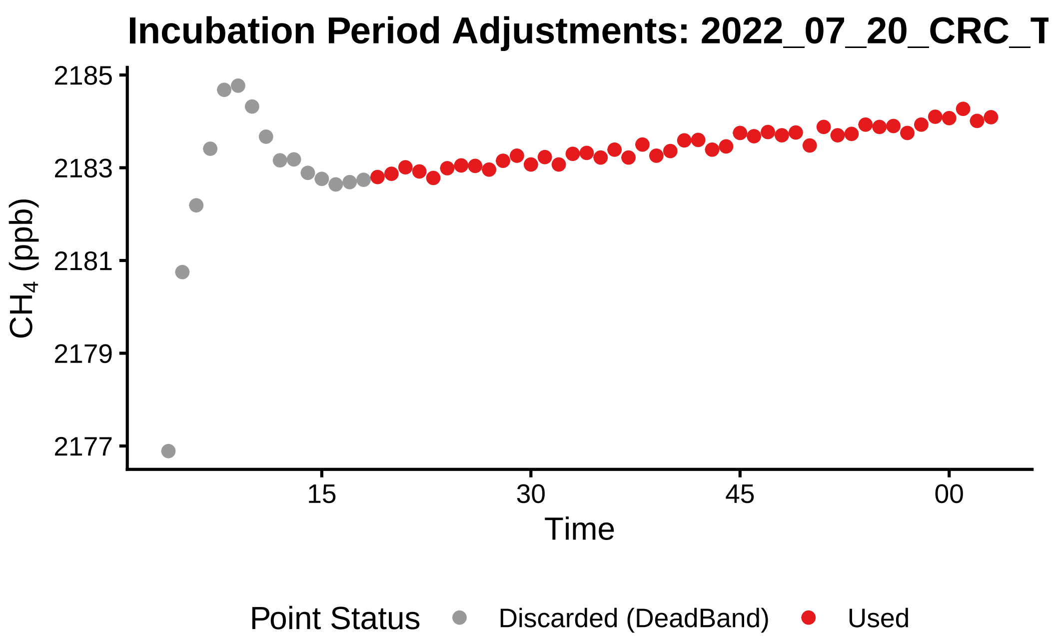
Incubation Period Adjustments: 2022_07_08_CRC_W
5000
4500
4000
3500
3000
CH4 (ppb)
13:25:30
13:26:00
13:26:30
13:27:00
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_07_08_CRC_T
2148
2146
2144
2142
2140
CH_4 (ppb)
14:12:30
14:13:00
14:13:30
14:14:00
Time
Point Status
Discarded (DeadBand)
Used

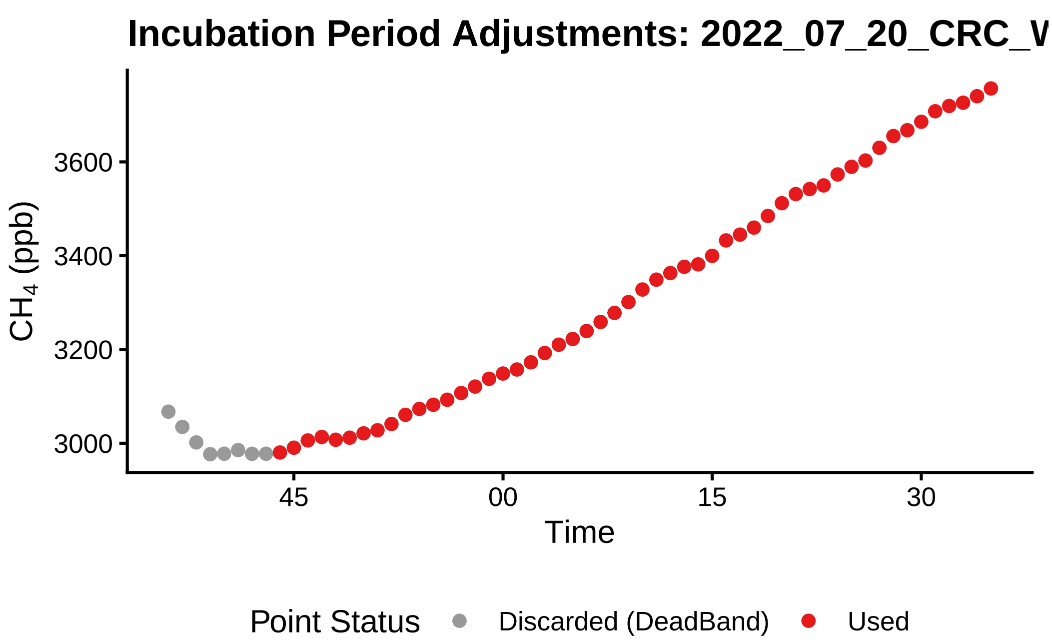

Incubation Period Adjustments: 2022_07_08_CRC_
2175.5
2175.0
2174.5
2174.0
2173.5
CH_4 (ppb)
14:29:00
14:29:30
14:30:00
14:30:30
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_10_14_CRC_
2102.4
2102.1
2101.8
2101.5
CH_4 (ppb)
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used

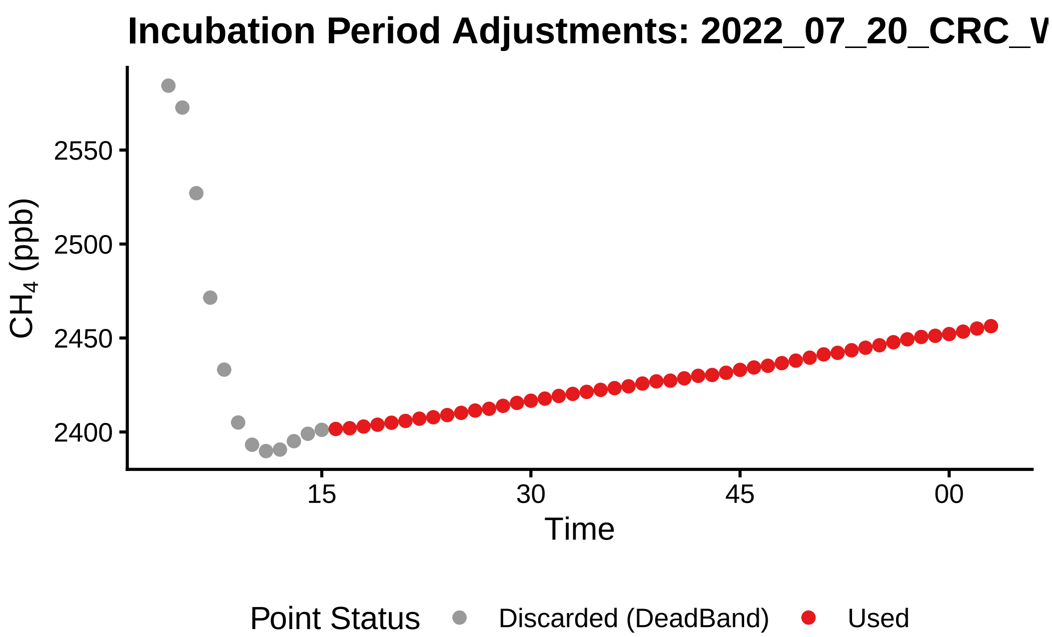

Incubation Period Adjustments: 2022_07_12_CRC_T
2055
2054
2053
2052
2051
CH_4 (ppb)
12:03:30
12:04:00
12:04:30
12:05:00
Time
Point Status
Discarded (DeadBand)
Used

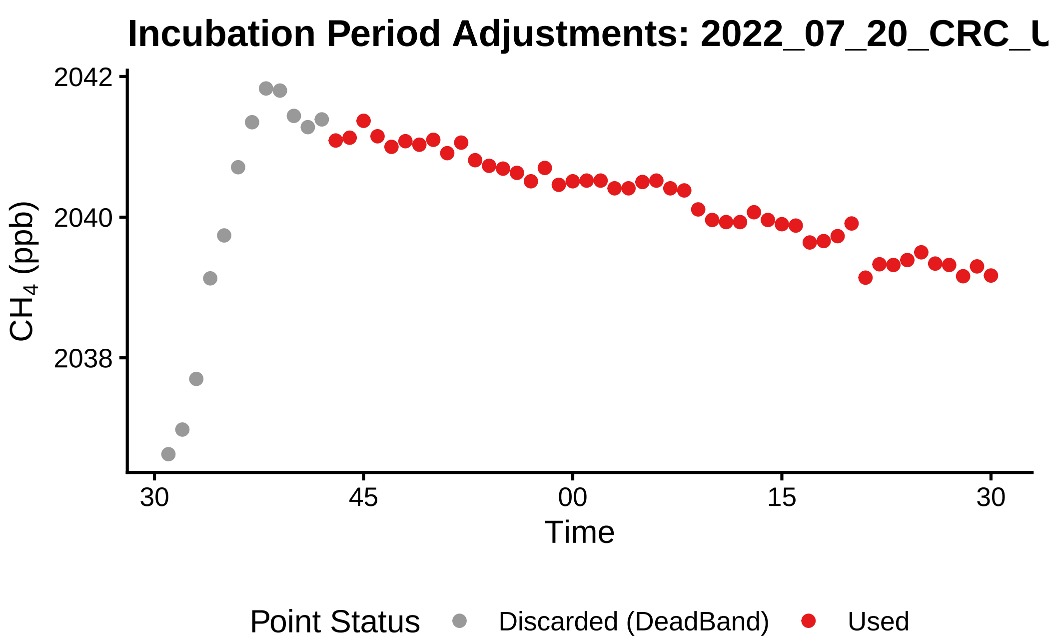


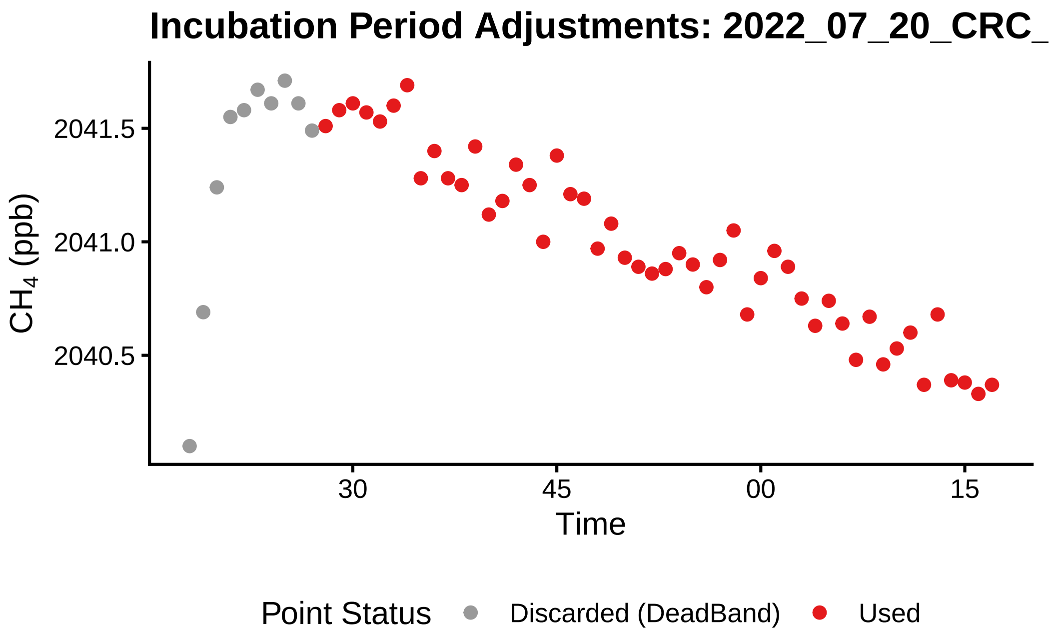
Incubation Period Adjustments: 2022_07_20_CRC_T
2185
2183
2181
2179
2177
CH4 (ppb)
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

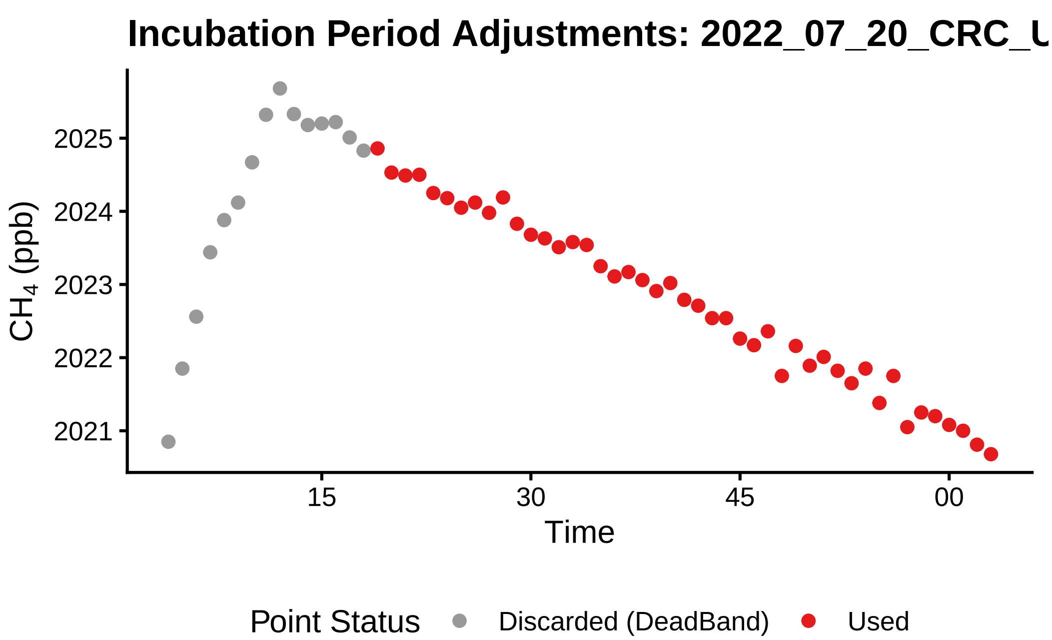


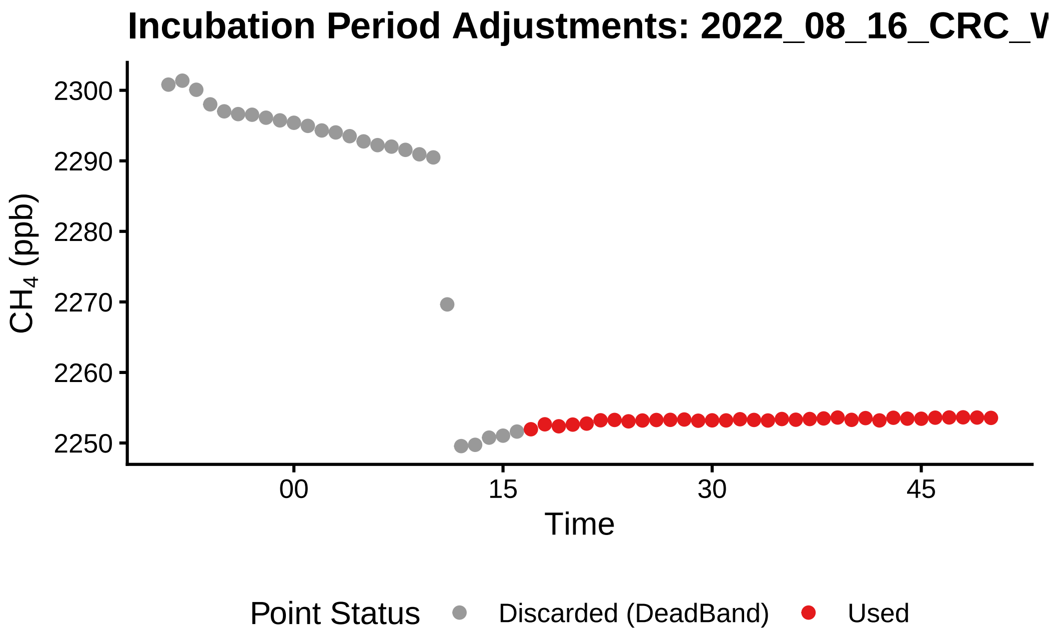
Incubation Period Adjustments: 2022_07_20_CRC_V
3600
3400
3200
3000
CH4 (ppb)
45
00
15
30
Time
Point Status
Discarded (DeadBand)
Used

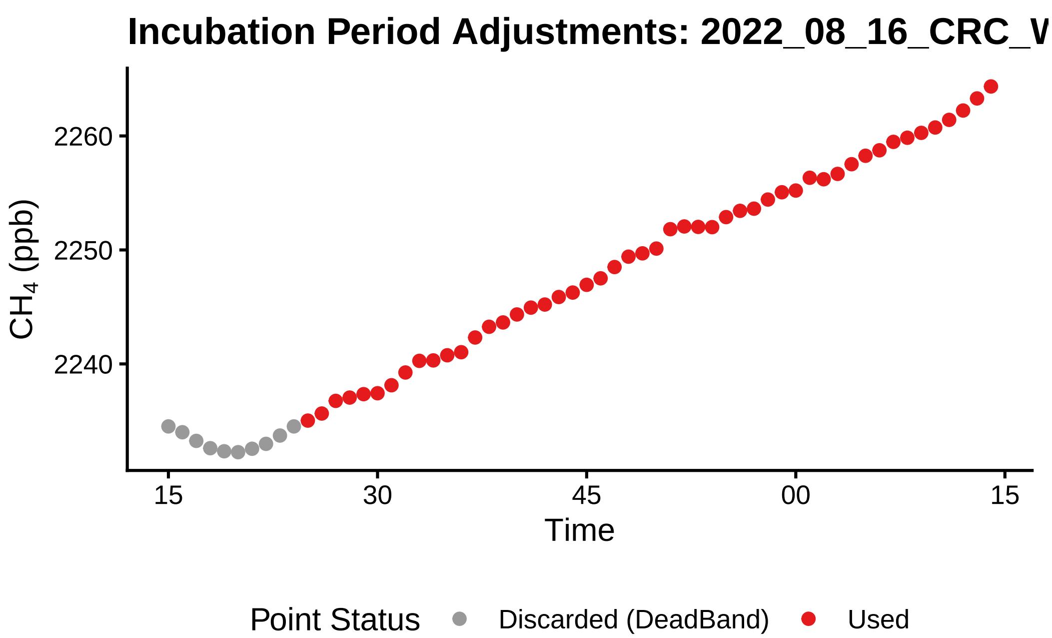

Incubation Period Adjustments: 2022_07_20_CRC_V
2550
2500
2450
2400
CH_4 (ppb)
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

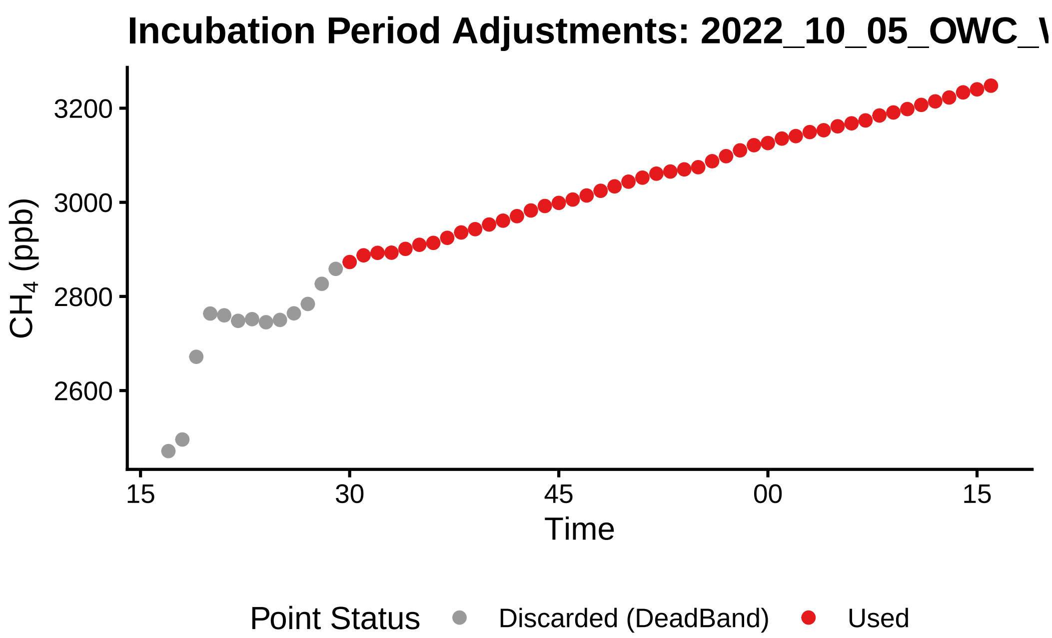

Incubation Period Adjustments: 2022_07_20_CRC_U
2042
2040
2038
CH_4 (ppb)
30
45
00
15
30
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_07_20_CRC_
CH4 (ppb)
2041.5
2041.0
2040.5
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used

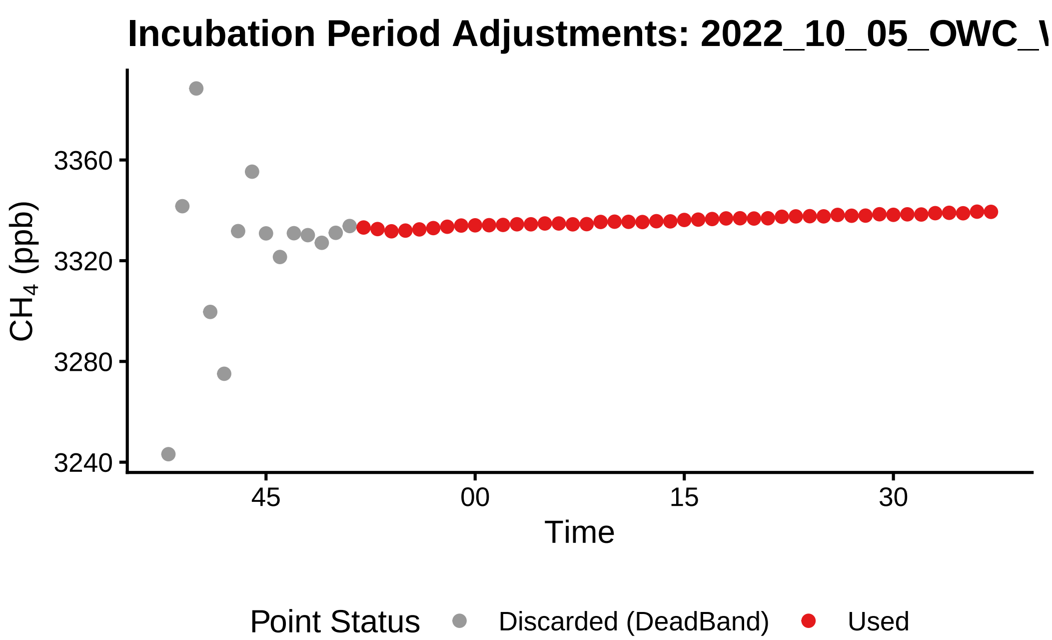

Incubation Period Adjustments: 2022_07_20_CRC_U
CH4 (ppb)
2025
2024
2023
2022
2021
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_08_16_CRC_V
CH4 (ppb)
2300
2290
2280
2270
2260
2250
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

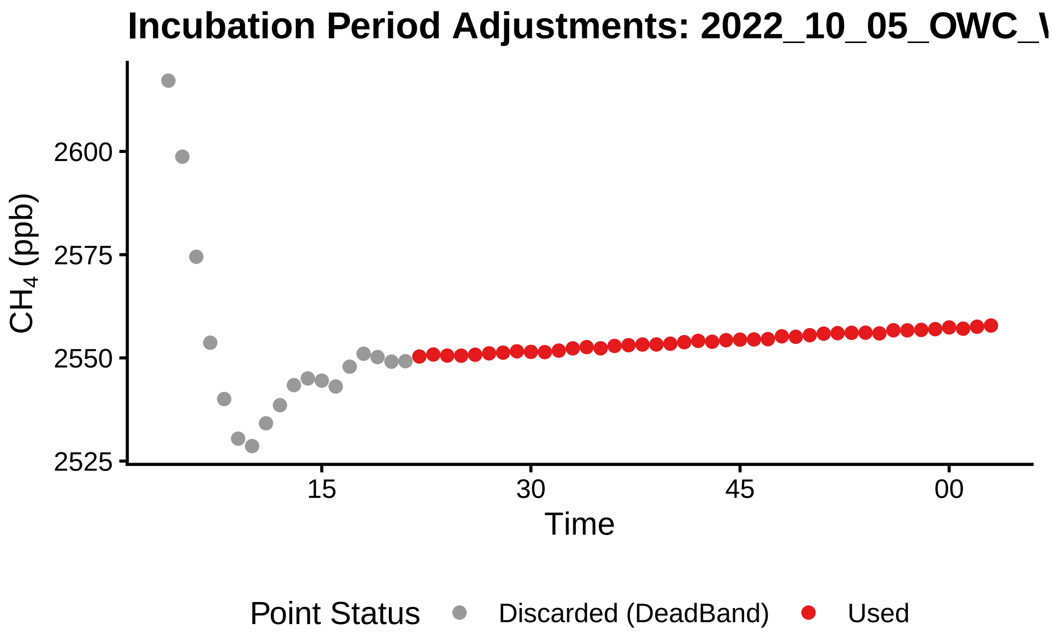

Incubation Period Adjustments: 2022_08_16_CRC_V
CH4 (ppb)
2260
2250
2240
15
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used

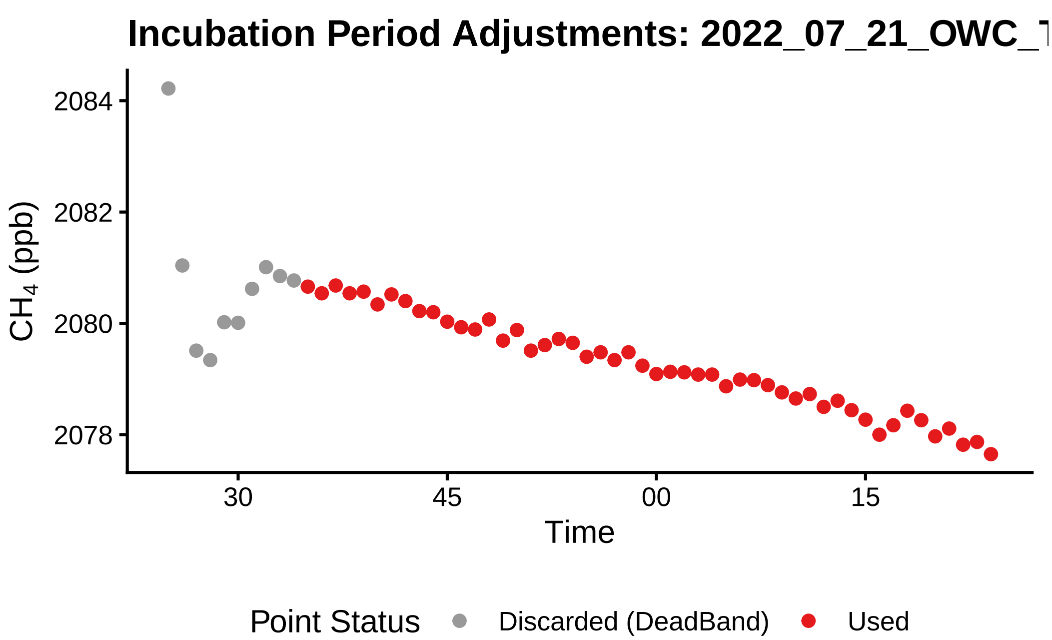


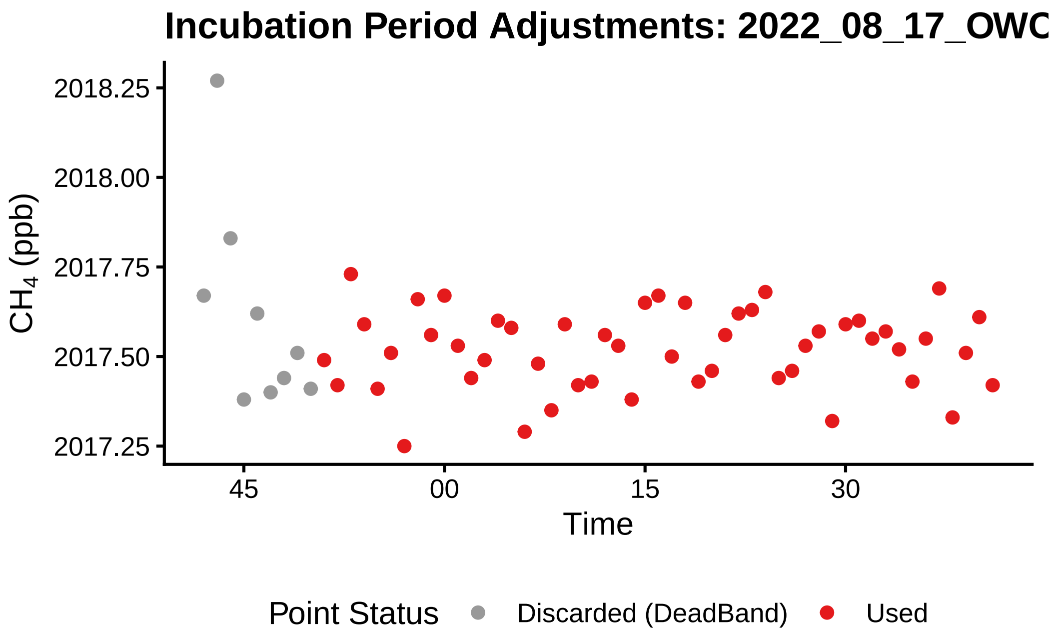
Incubation Period Adjustments: 2022_10_05_OWC_\
CH_4 (ppb)
3200
3000
2800
2600
15
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used

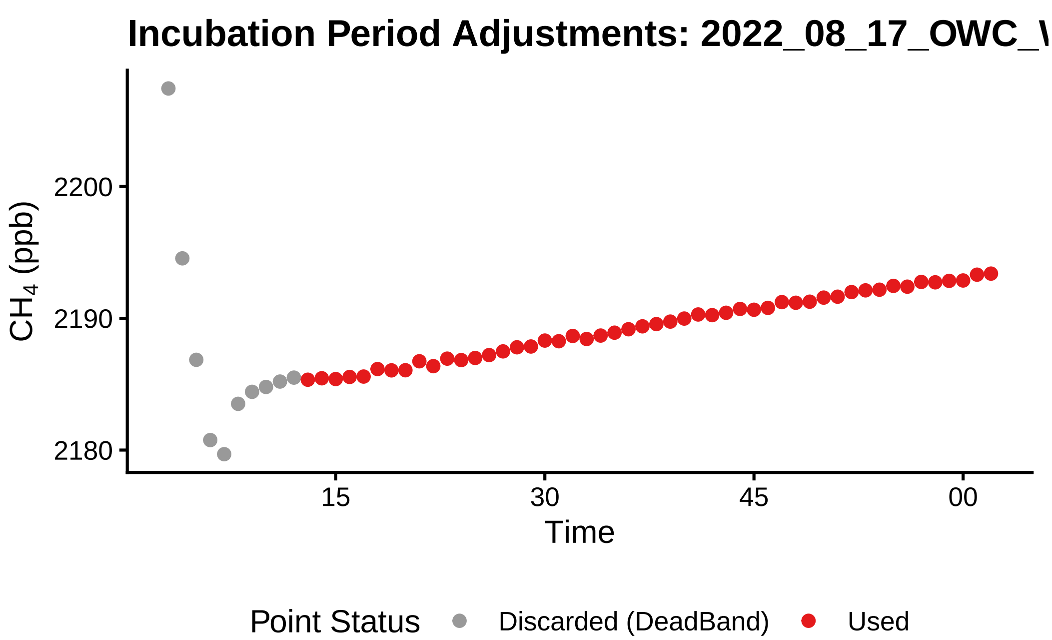


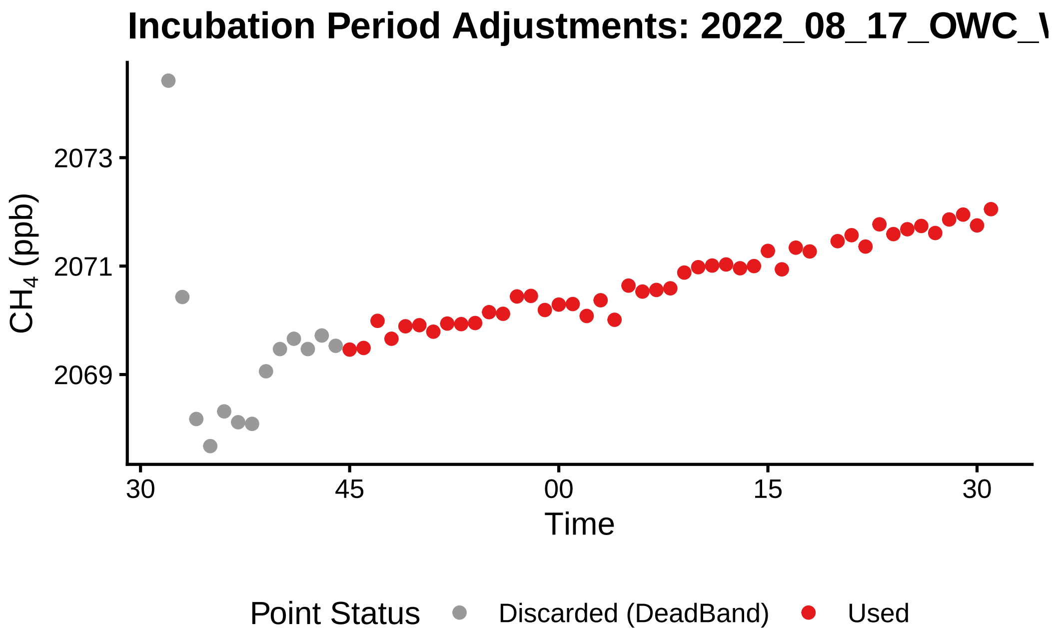
Incubation Period Adjustments: 2022_10_05_OWC_\
CH_4 (ppb)
3360
3320
3280
3240
45
00
15
30
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_10_05_OWC_\
CH4 (ppb)
2600
2575
2550
2525
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used

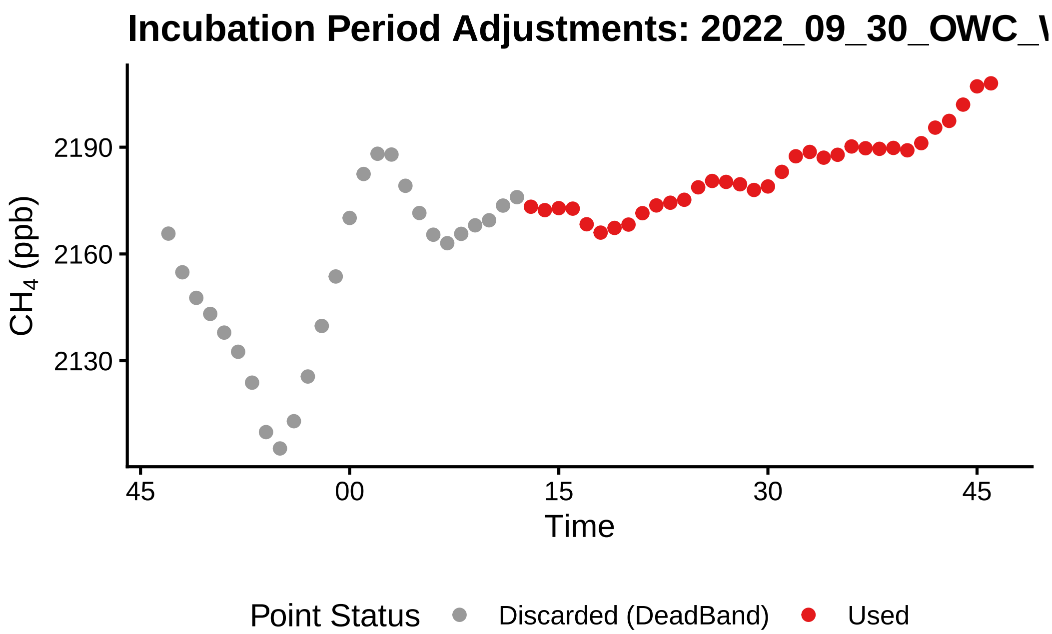

Incubation Period Adjustments: 2022_07_21_OWC_T
CH4 (ppb)
2084
2082
2080
2078
30
45
00
15
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_08_17_OWC
2018.25
2018.00
2017.75
2017.50
2017.25
CH4 (ppb)
45
00
15
30
Time
Point Status
Discarded (DeadBand)
Used

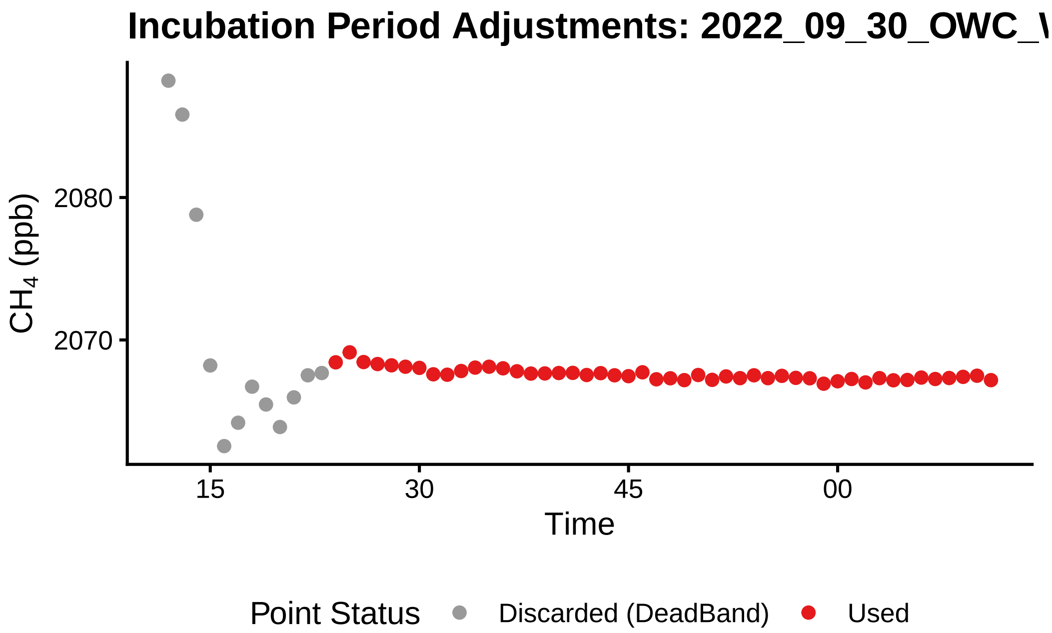

Incubation Period Adjustments: 2022_08_17_OWC_\
2200
2190
2180
CH4 (ppb)
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used



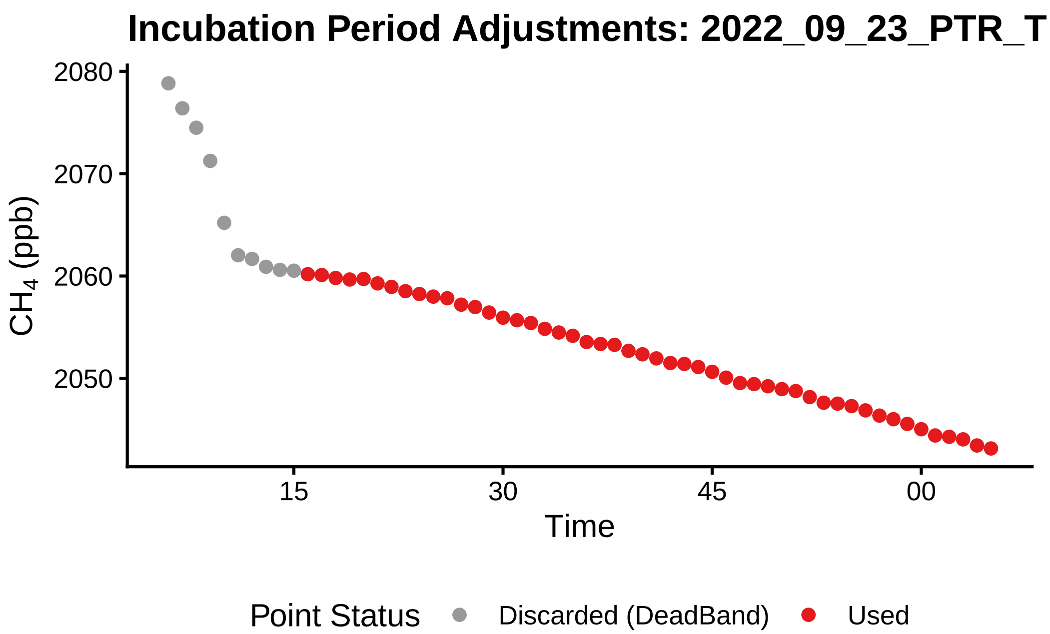
Incubation Period Adjustments: 2022_08_17_OWC_\
2073
2071
2069
CH4 (ppb)
30
45
00
15
30
Time
Point Status
Discarded (DeadBand)
Used



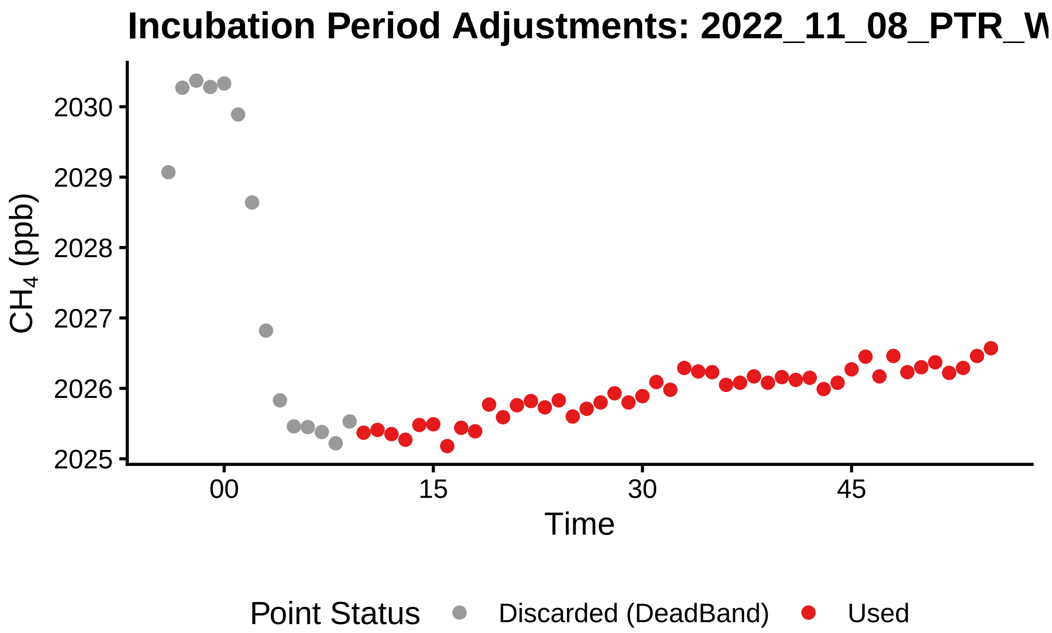
Incubation Period Adjustments: 2022_09_30_OWC_\
2190
2160
2130
CH4 (ppb)
45
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

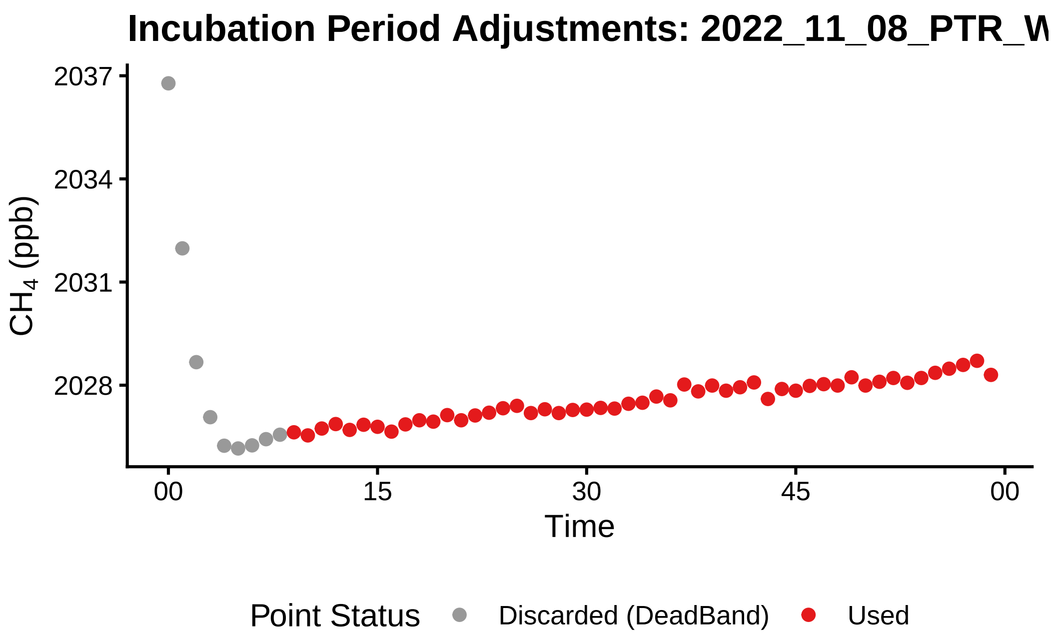


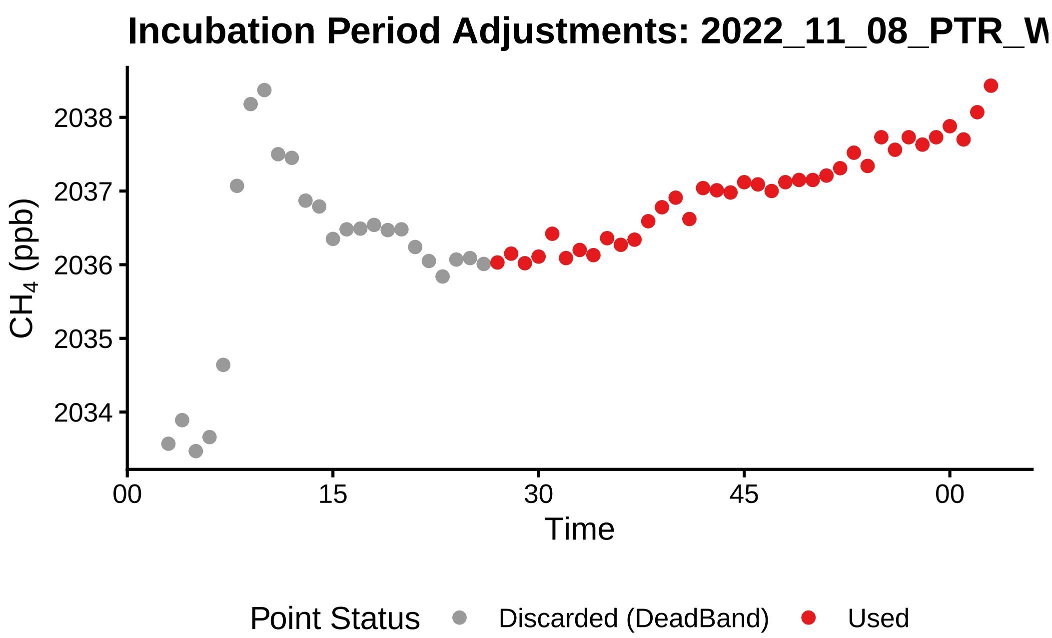
Incubation Period Adjustments: 2022_09_30_OWC_\
2080
2070
CH4 (ppb)
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used



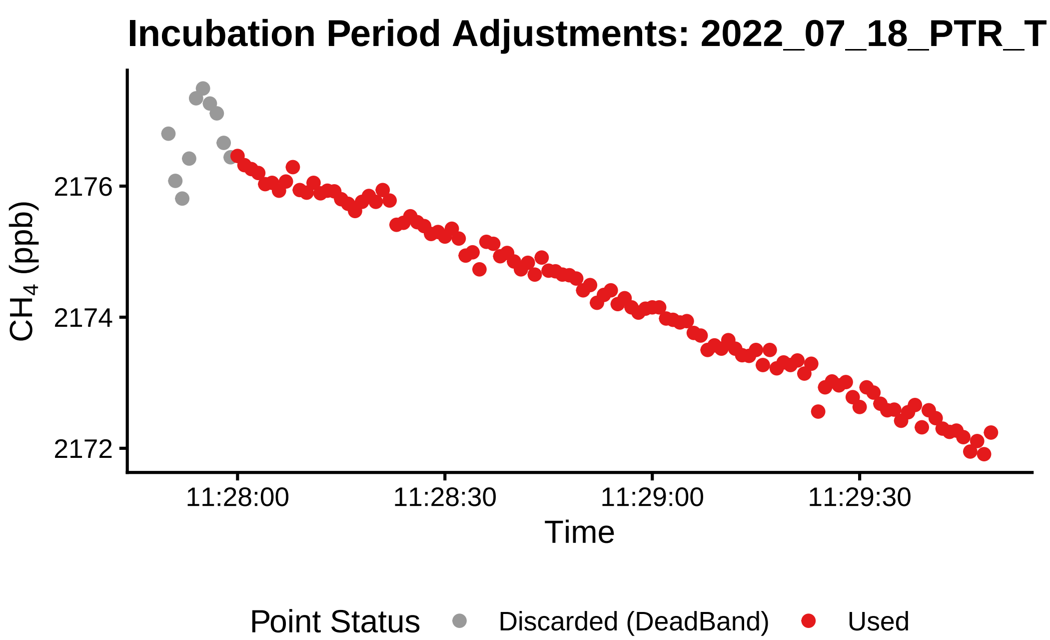
Incubation Period Adjustments: 2022_09_23_PTR_T
2080
2070
2060
2050
CH4 (ppb)
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_11_08_PTR_W
CH4 (ppb)
2030
2029
2028
2027
2026
2025
00
15
30
45
Time
Point Status
Discarded (DeadBand)
Used

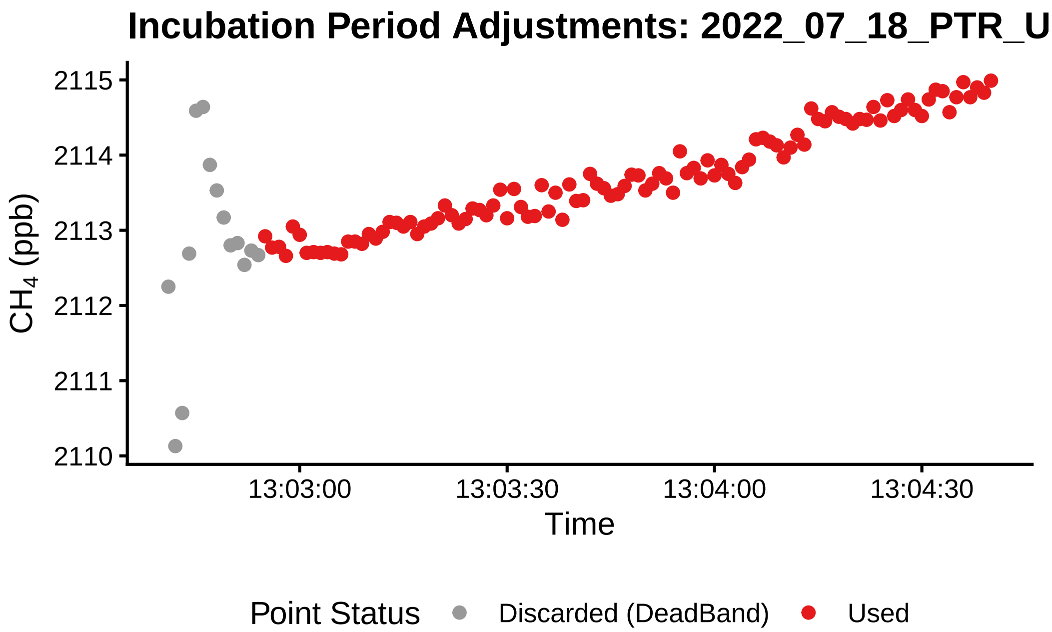

Incubation Period Adjustments: 2022_11_08_PTR_W
CH4 (ppb)
2037
2034
2031
2028
00
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used



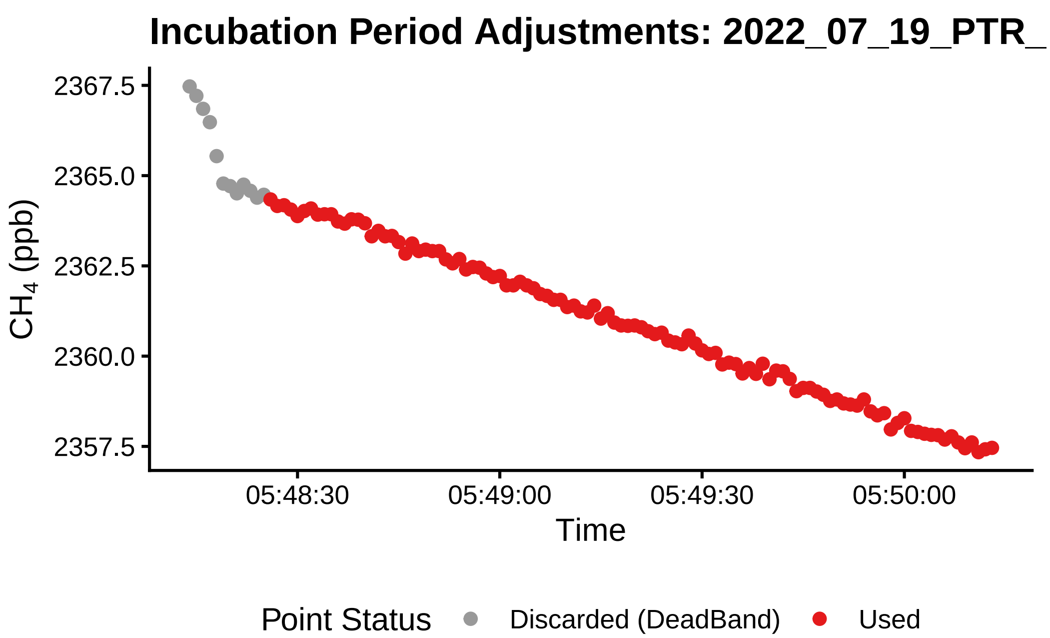
Incubation Period Adjustments: 2022_11_08_PTR_W
2038
2037
2036
2035
2034
CH4 (ppb)
00
15
30
45
00
Time
Point Status
Discarded (DeadBand)
Used



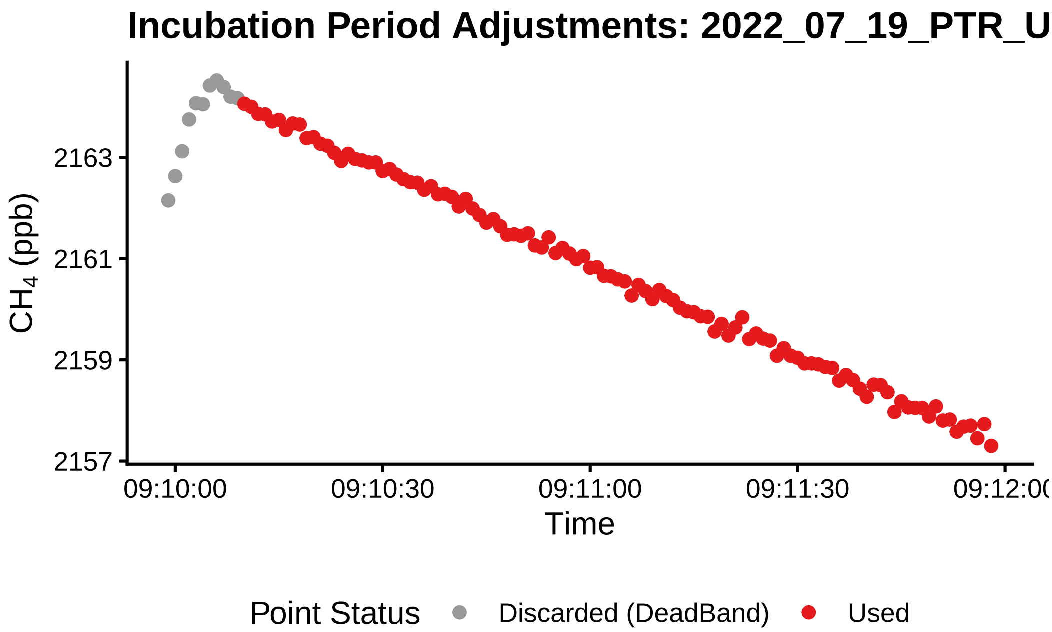
Incubation Period Adjustments: 2022_07_18_PTR_T
2176
2174
2172
CH4 (ppb)
11:28:00
11:28:30
11:29:00
11:29:30
Time
Point Status
Discarded (DeadBand)
Used

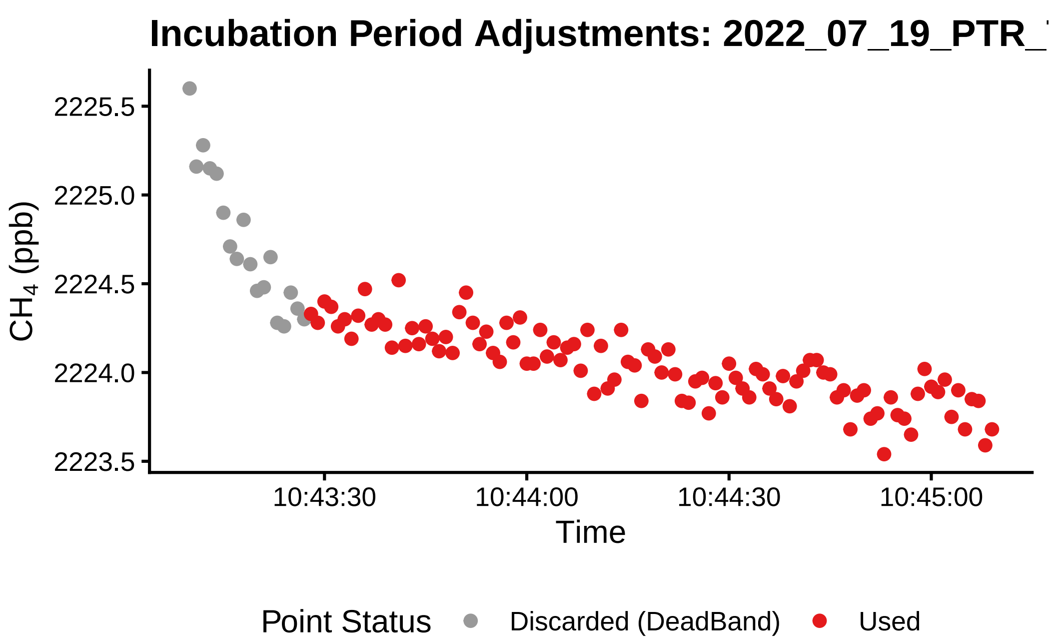


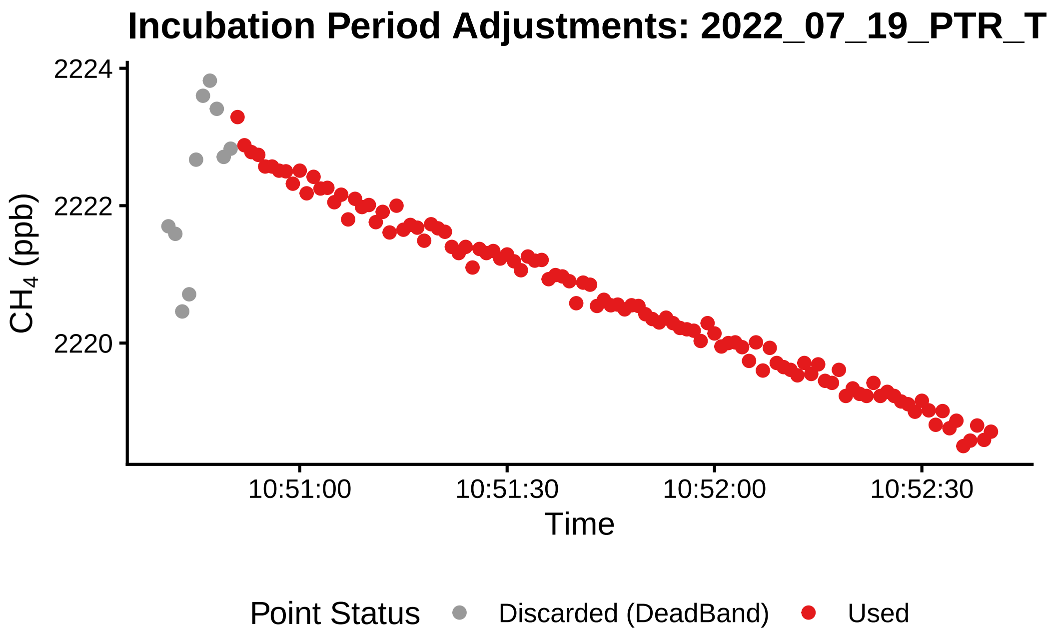
Incubation Period Adjustments: 2022_07_18_PTR_U
2115
2114
2113
2112
2111
2110
CH4 (ppb)
13:03:00
13:03:30
13:04:00
13:04:30
Time
Point Status
Discarded (DeadBand)
Used

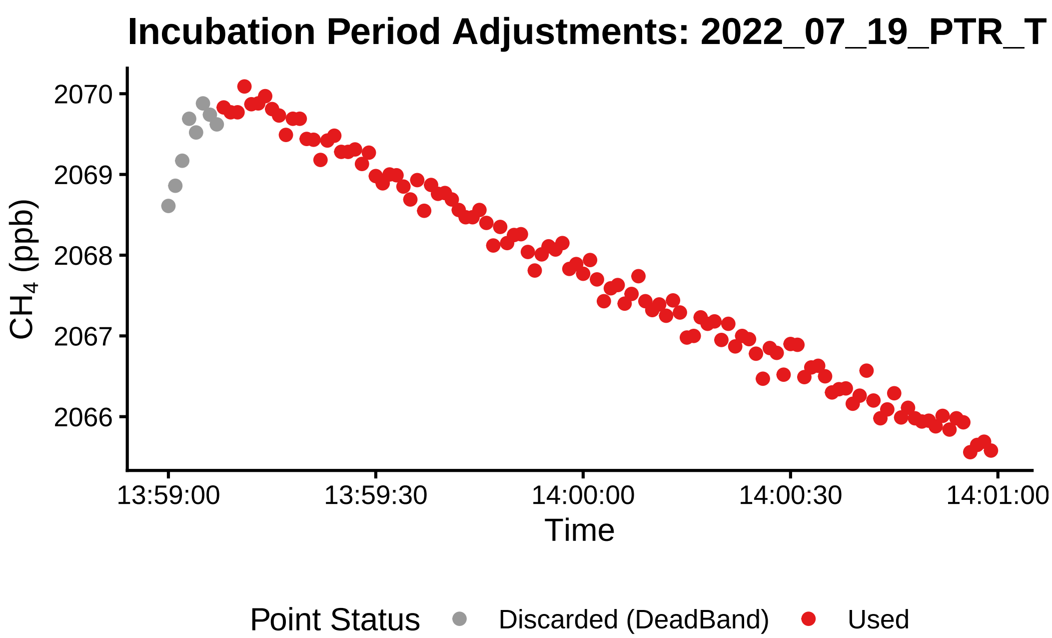


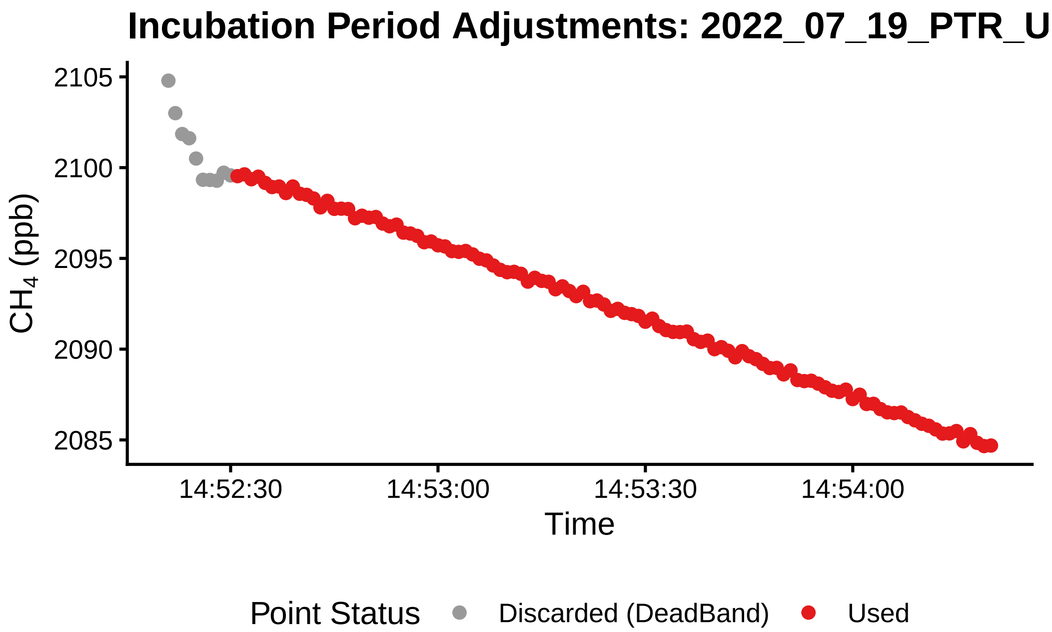
Incubation Period Adjustments: 2022_07_19_PTR_
2367.5
2365.0
2362.5
2360.0
2357.5
CH4 (ppb)
05:48:30
05:49:00
05:49:30
05:50:00
Time
Point Status
Discarded (DeadBand)
Used

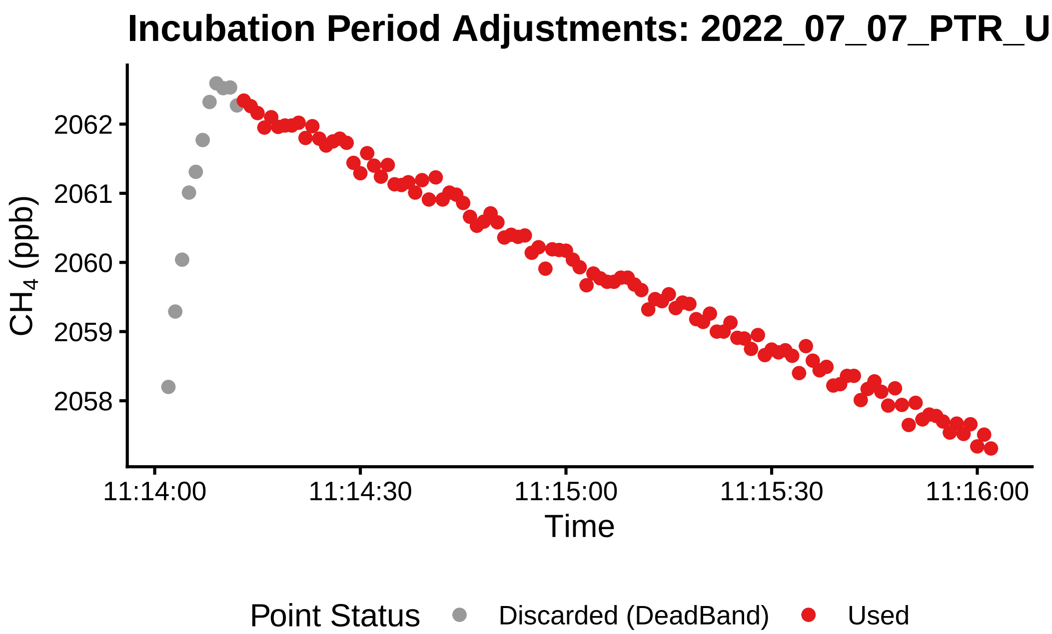

Incubation Period Adjustments: 2022_07_19_PTR_U
CH4 (ppb)
2163
2161
2159
2157
09:10:00
09:10:30
09:11:00
09:11:30
Time
Point Status
Discarded (DeadBand)
Used

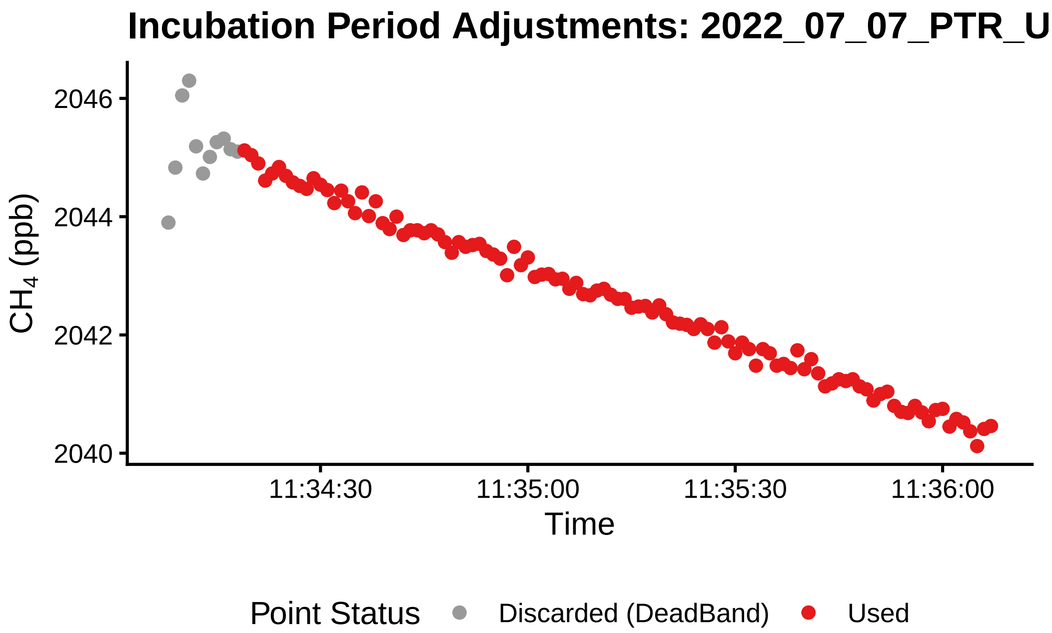

Incubation Period Adjustments: 2022_07_19_PTR_
CH4 (ppb)
2225.5
2225.0
2224.5
2224.0
2223.5
10:43:30
10:44:00
10:44:30
10:45:00
Time
Point Status
Discarded (DeadBand)
Used

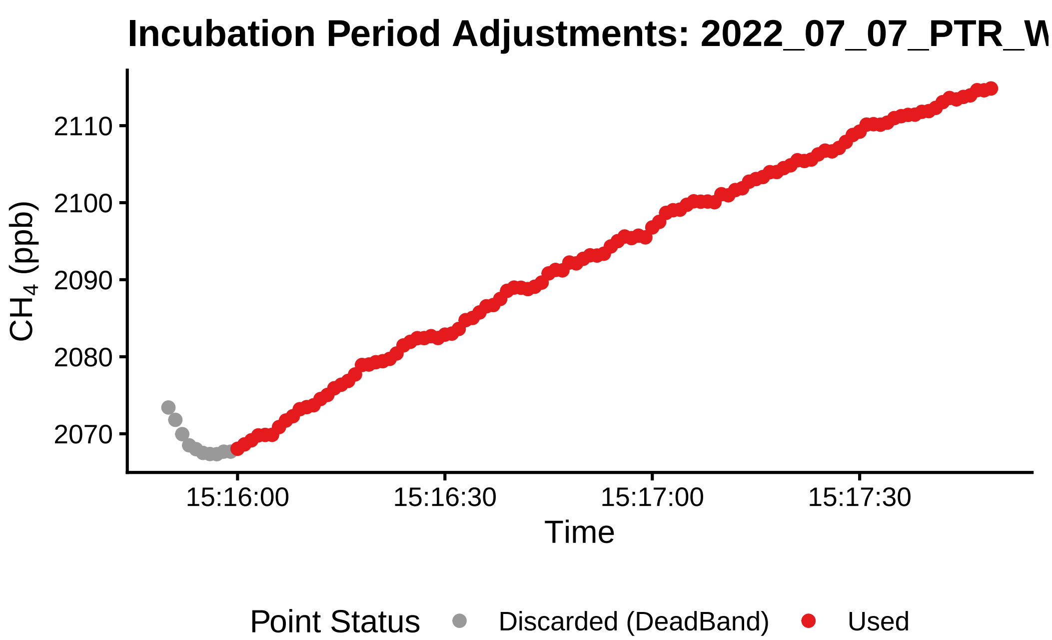

Incubation Period Adjustments: 2022_07_19_PTR_T
2224
2222
2220
CH4 (ppb)
10:51:00
10:51:30
10:52:00
10:52:30
Time
Point Status
Discarded (DeadBand)
Used

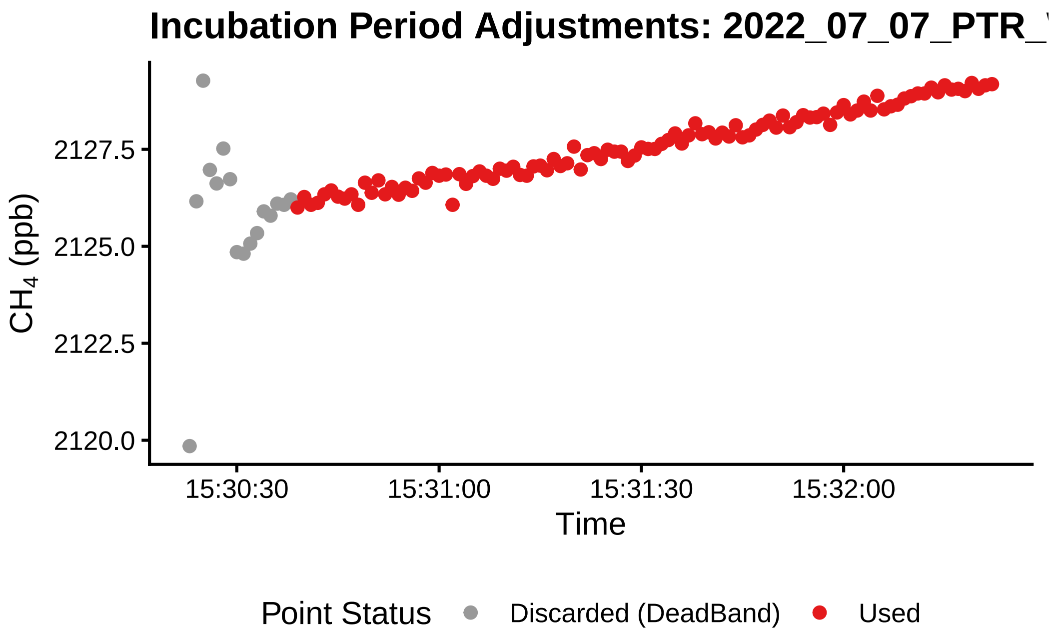

Incubation Period Adjustments: 2022_07_19_PTR_T
2070
2069
2068
2067
2066
CH4 (ppb)
13:59:00
13:59:30
14:00:00
14:00:30
14:01:00
Time
Point Status
Discarded (DeadBand)
Used

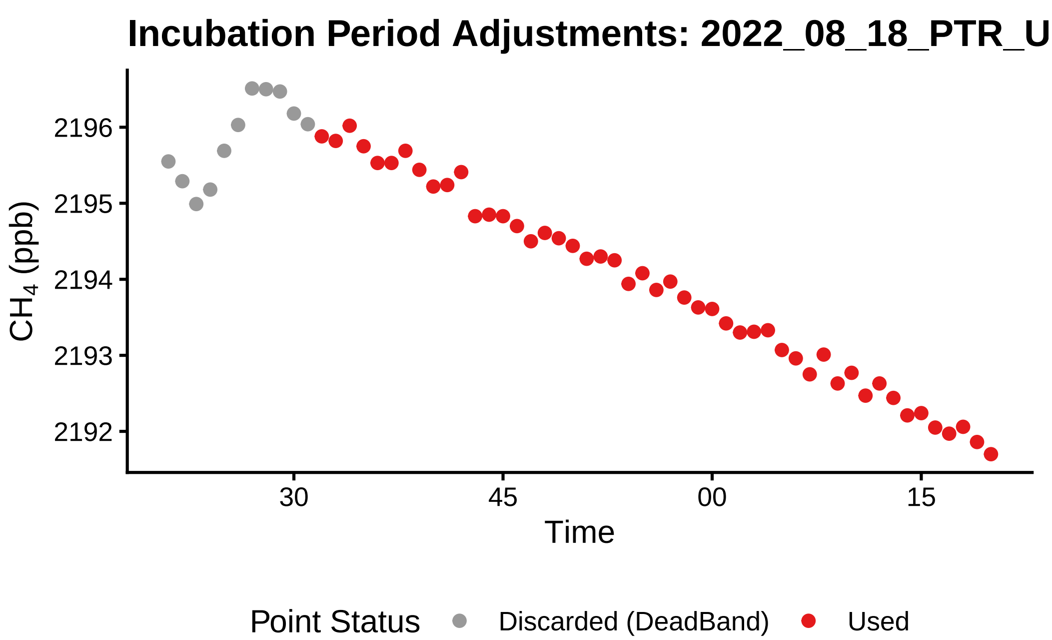

Incubation Period Adjustments: 2022_07_19_PTR_U
CH4 (ppb)
2105
2100
2095
2090
2085
14:52:30
14:53:00
14:53:30
14:54:00
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_07_07_PTR_U
CH4 (ppb)
2062
2061
2060
2059
2058
11:14:00
11:14:30
11:15:00
11:15:30
11:16:00
Time
Point Status
Discarded (DeadBand)
Used


Incubation Period Adjustments: 2022_07_07_PTR_U
CH4 (ppb)
2046
2044
2042
2040
11:34:30
11:35:00
11:35:30
11:36:00
Time
Point Status
Discarded (DeadBand)
Used

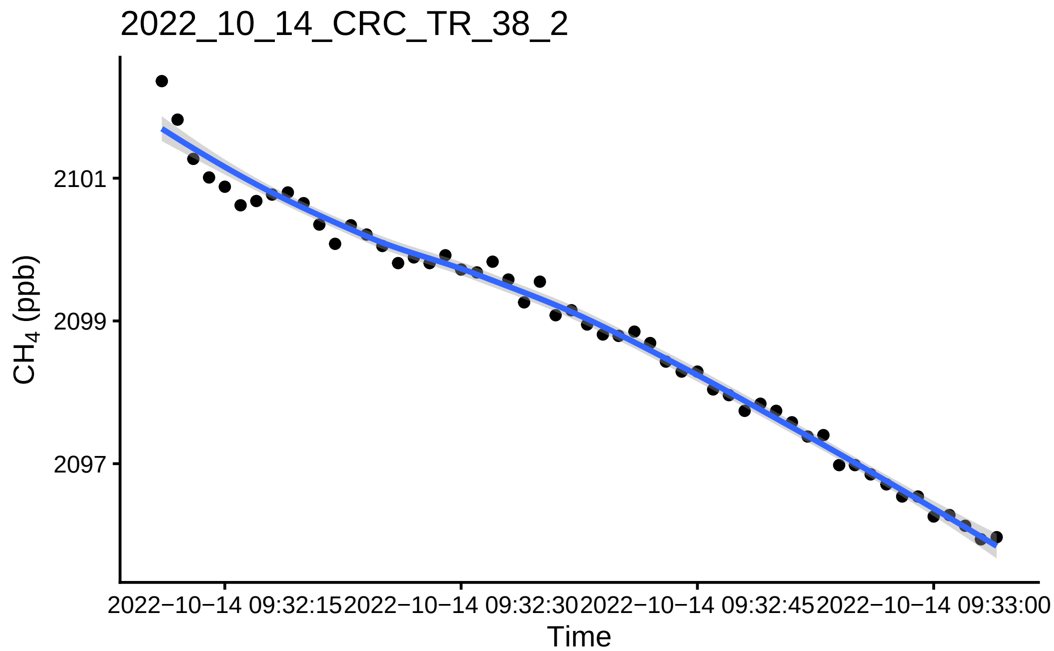

Incubation Period Adjustments: 2022_07_07_PTR_W
CH4 (ppb)
2110
2100
2090
2080
2070
15:16:00
15:16:30
15:17:00
15:17:30
Time
Point Status
Discarded (DeadBand)
Used



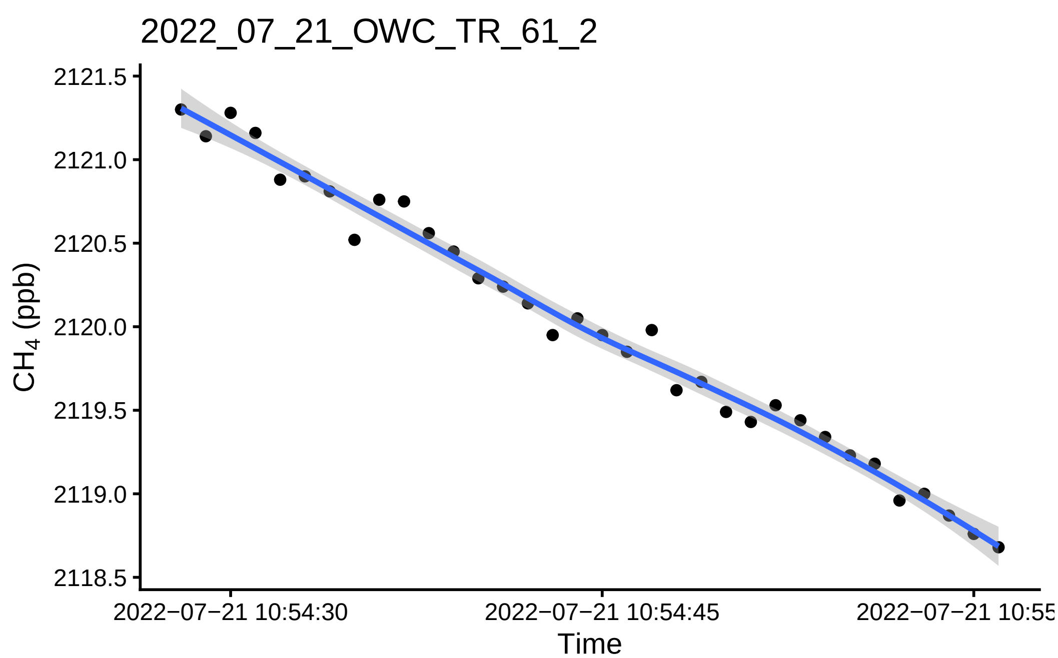


0.13 Spot check a few fluxes with R2 > 0.9

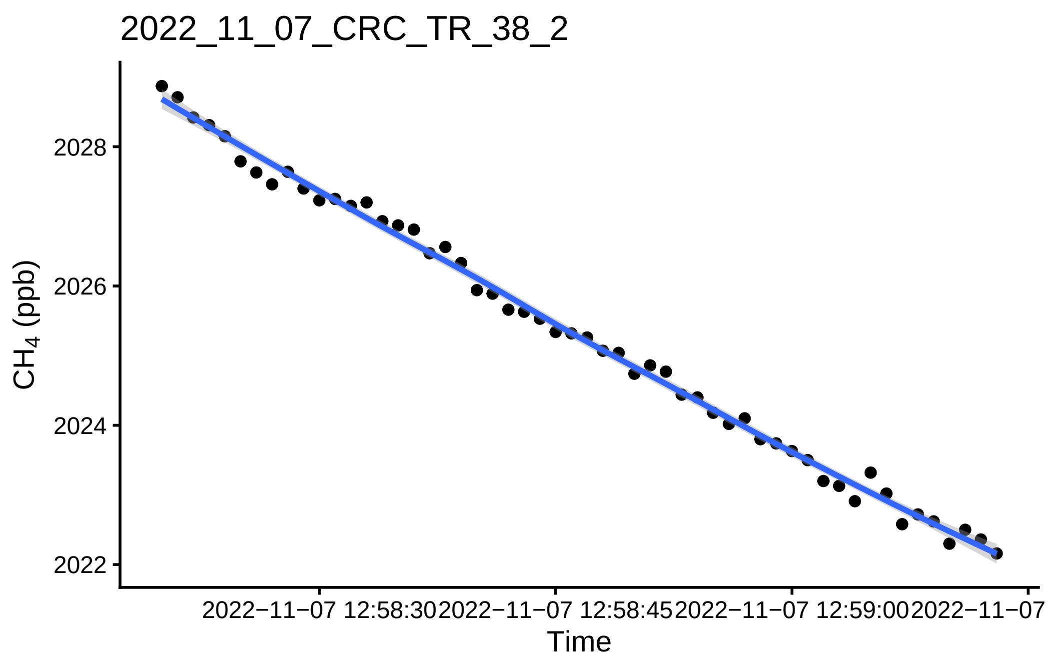
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



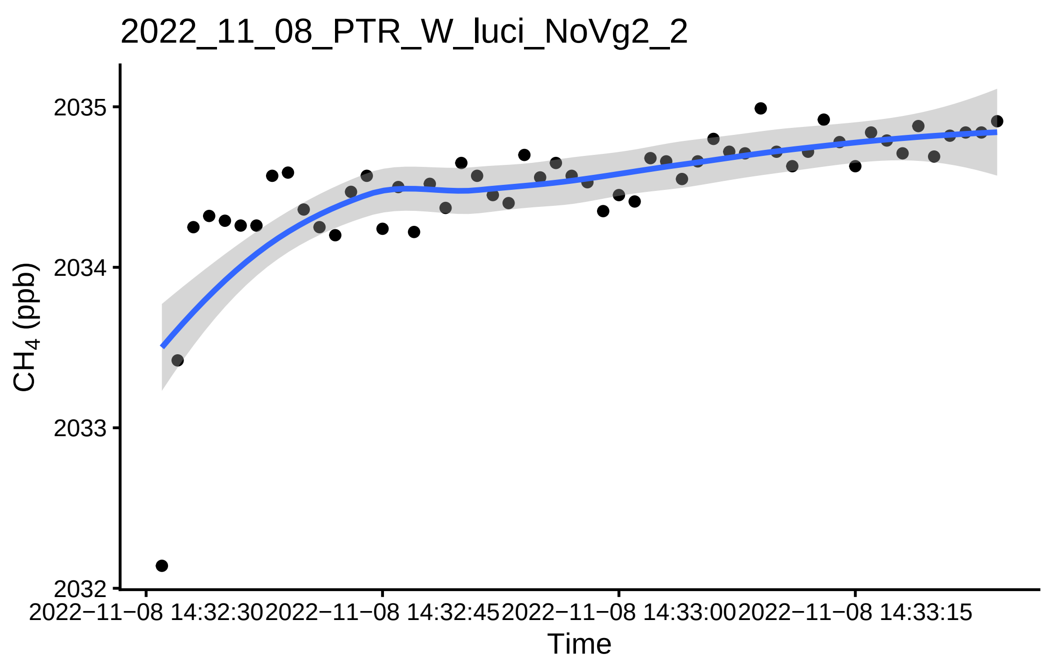

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



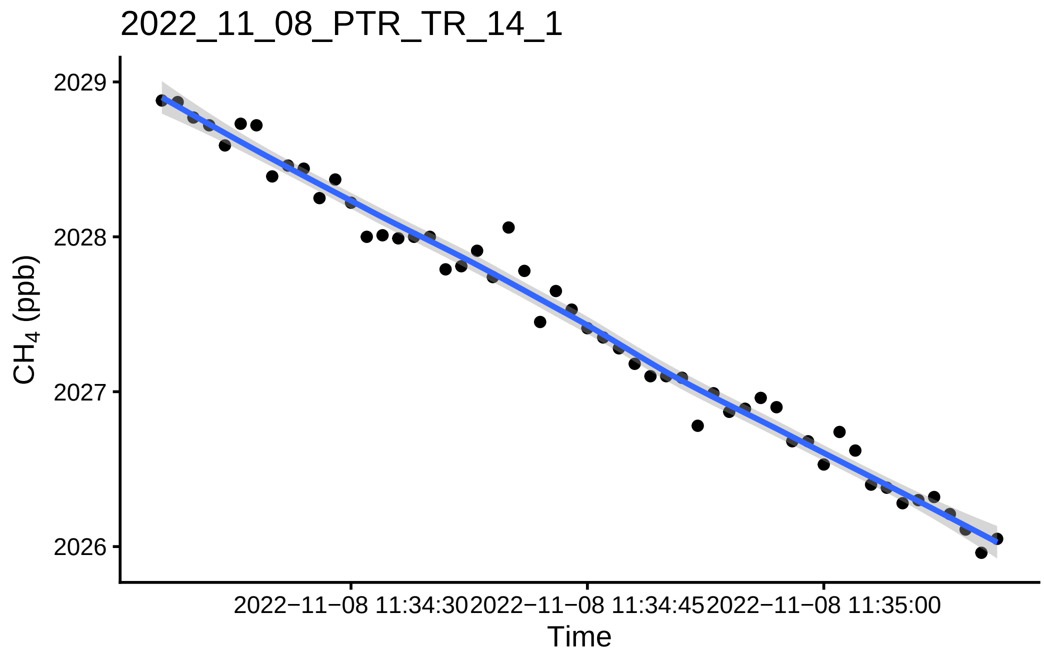

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

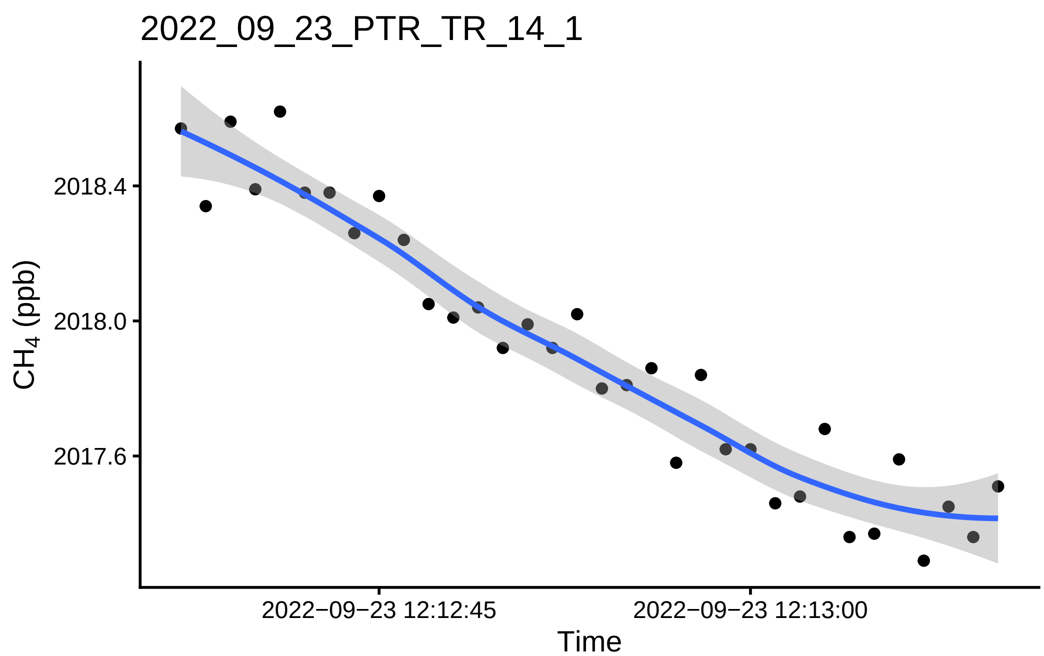



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

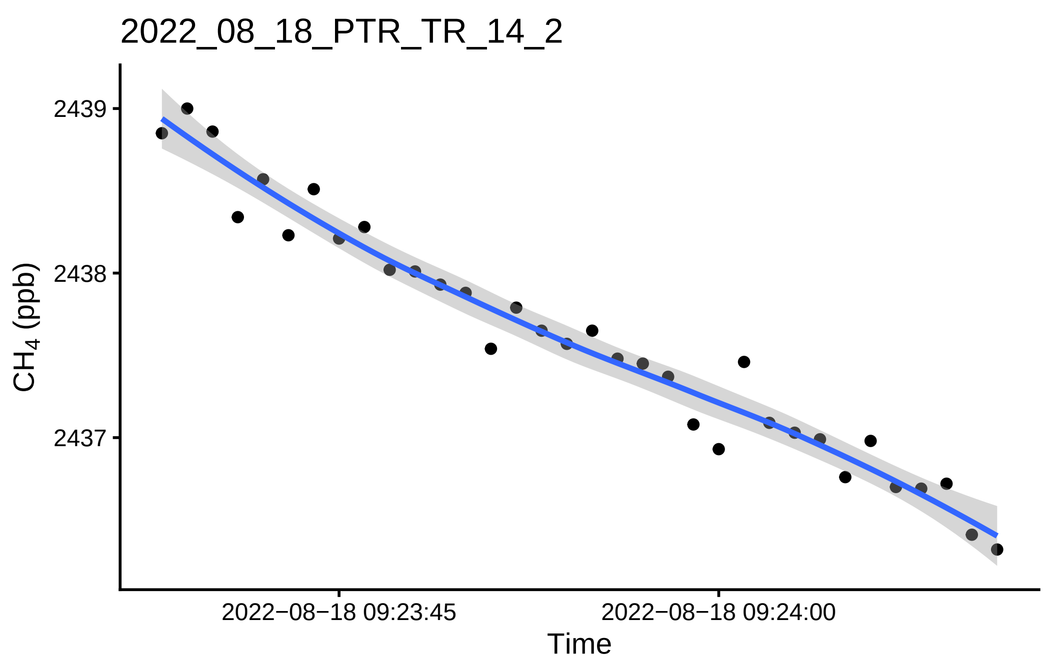


0.14 Take a look at the fluxes <0.9 and a flux estimate below 1

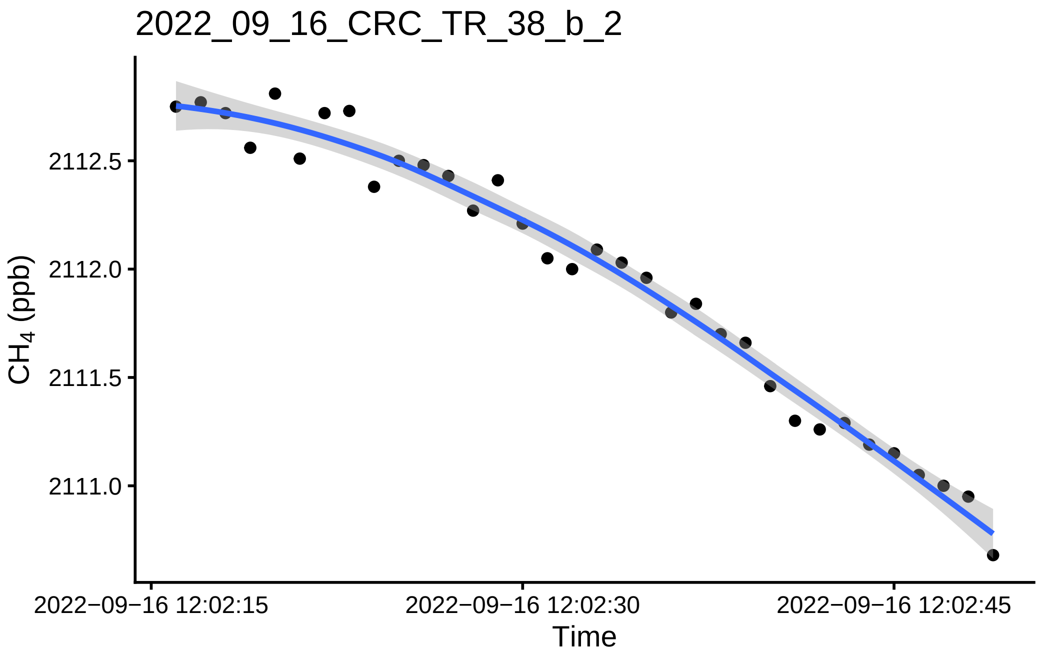
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



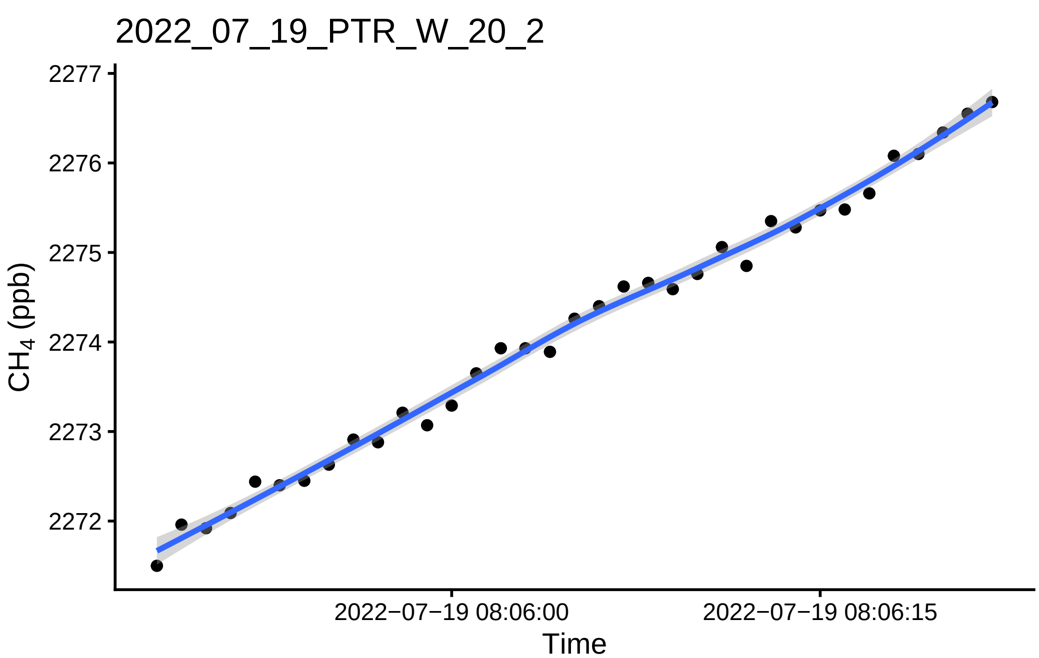

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



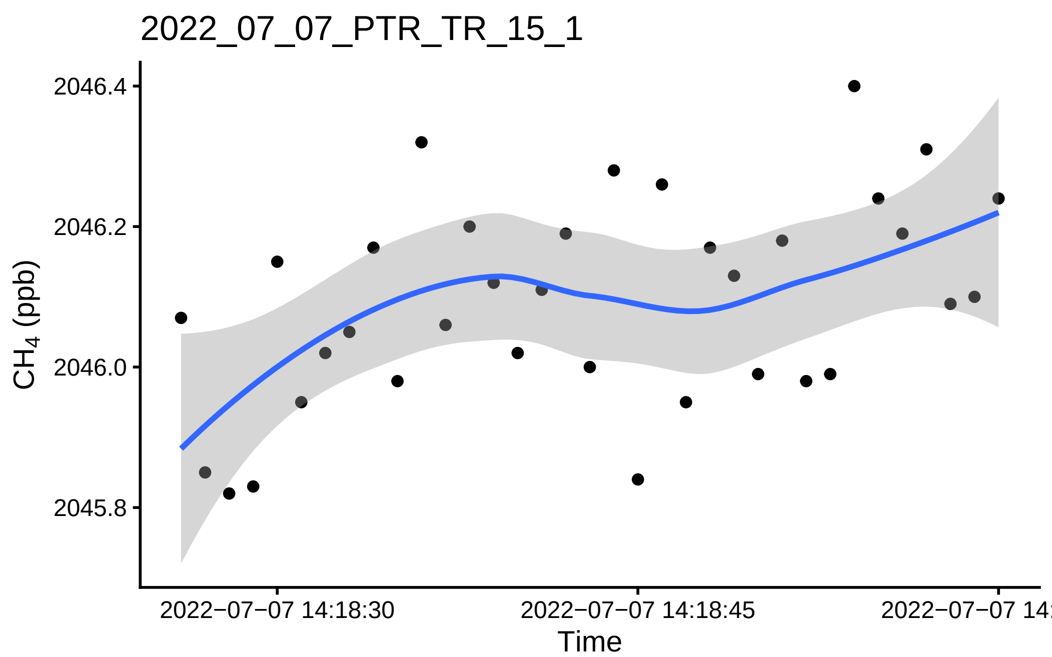

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



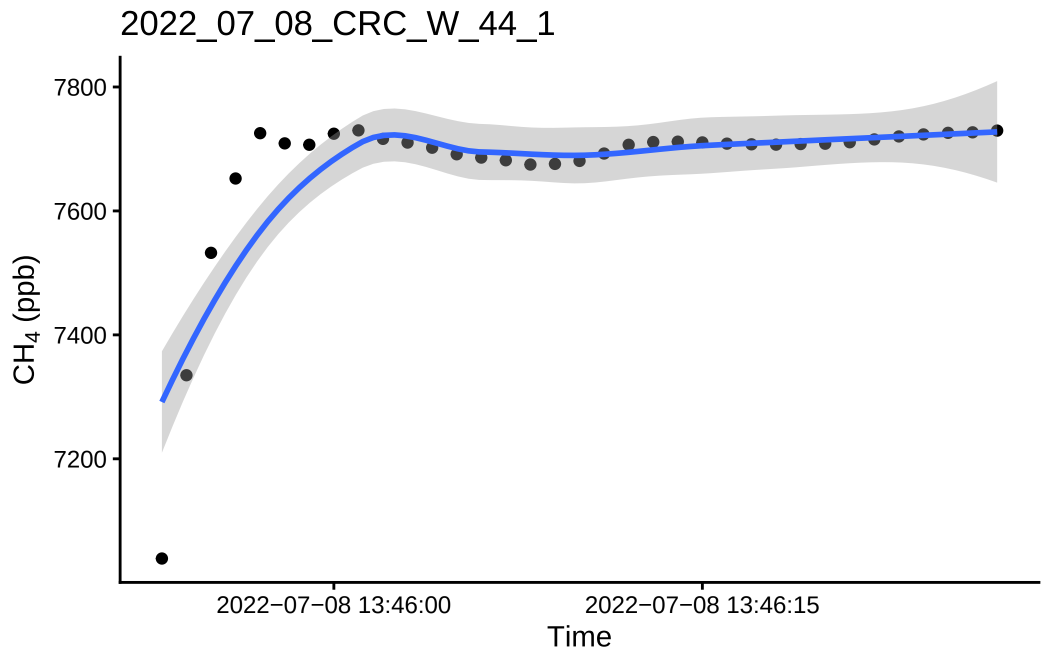

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



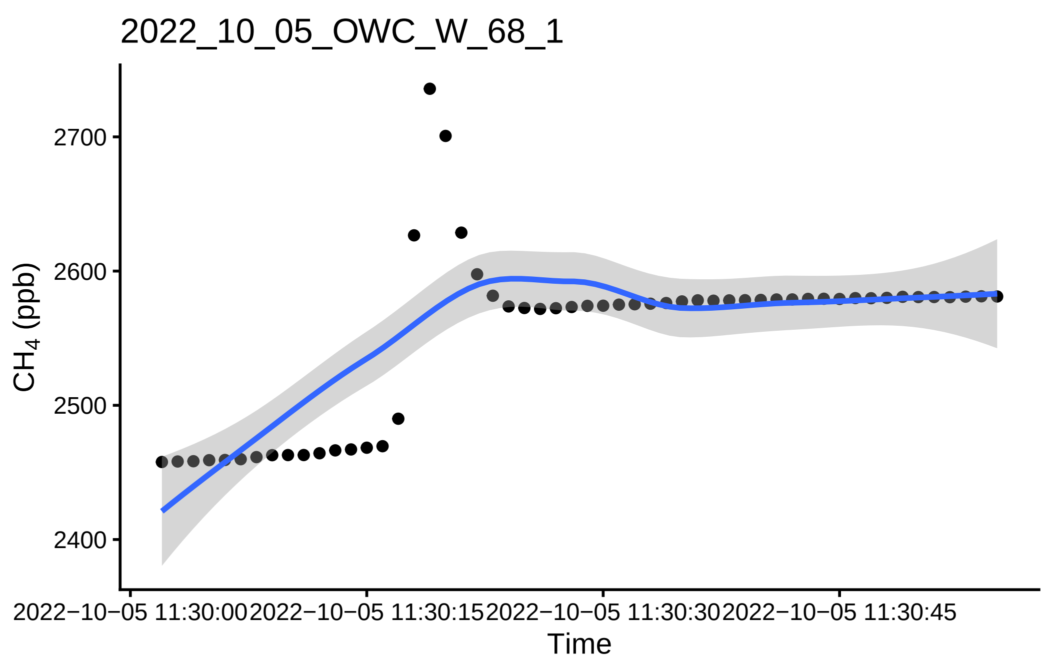
0.15 Take a look at the bad fluxes <0.9 and have a high flux estimate

1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

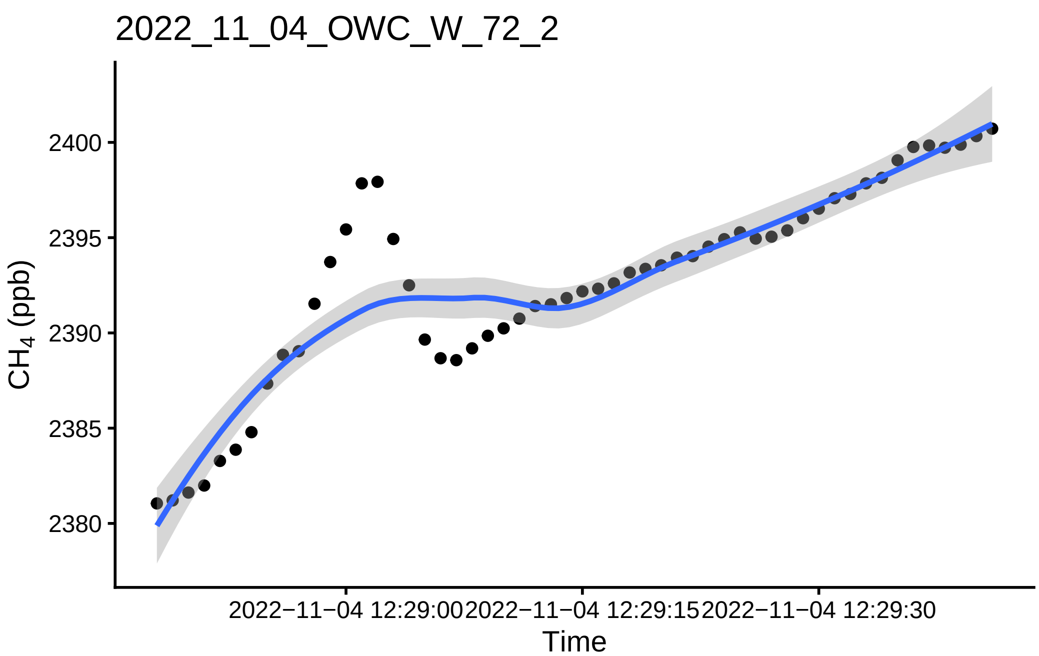
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



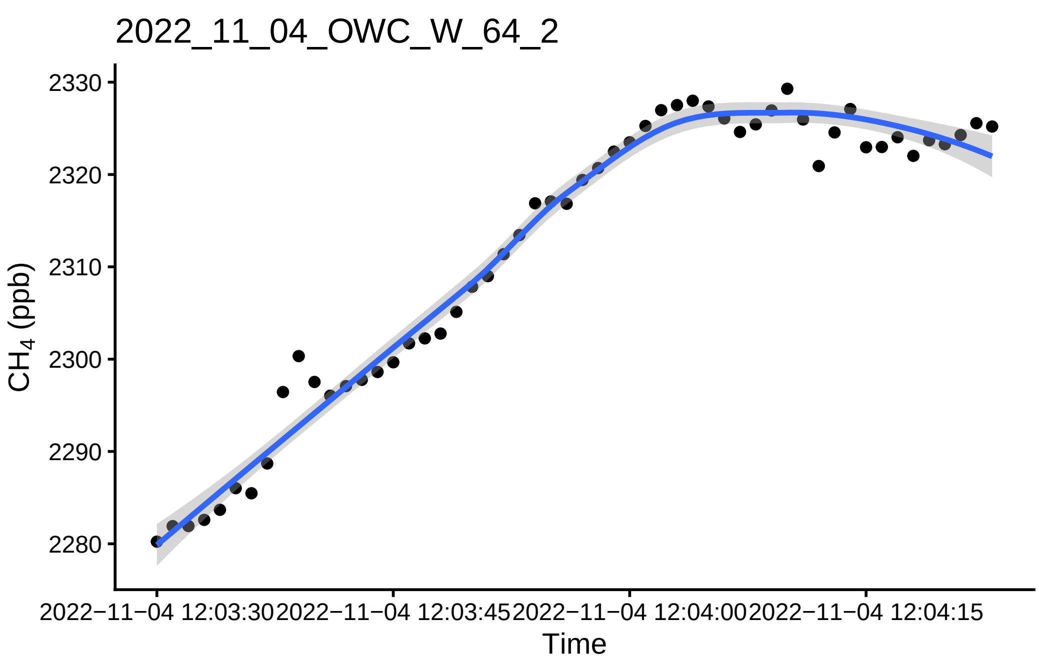

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



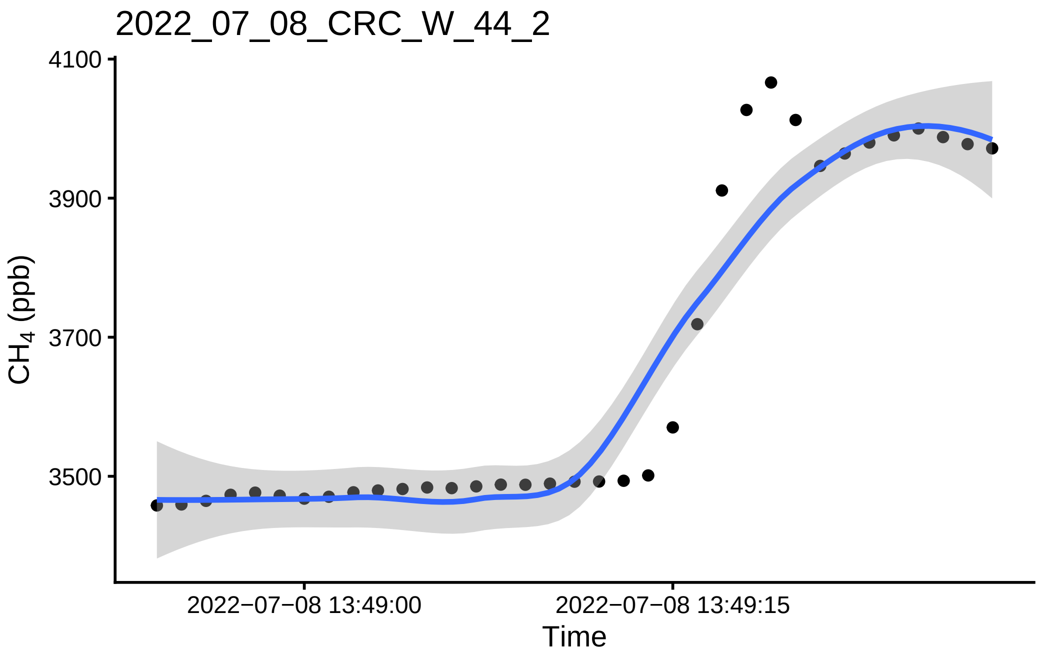

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

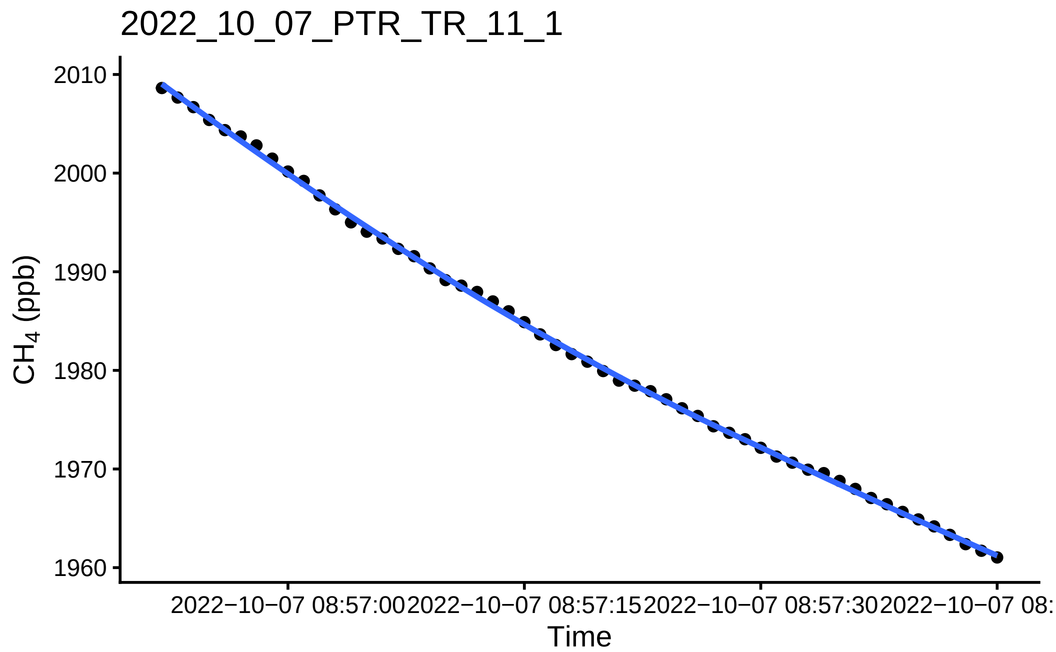



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

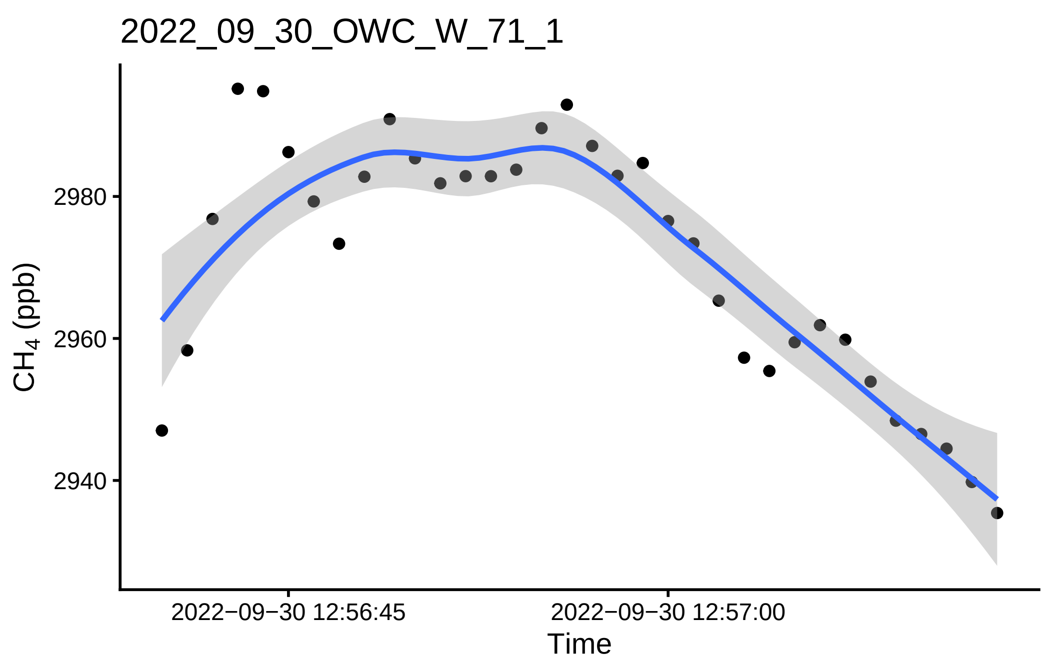


1.1 Look at the really high or really low ones to see if there is a timing issue or something

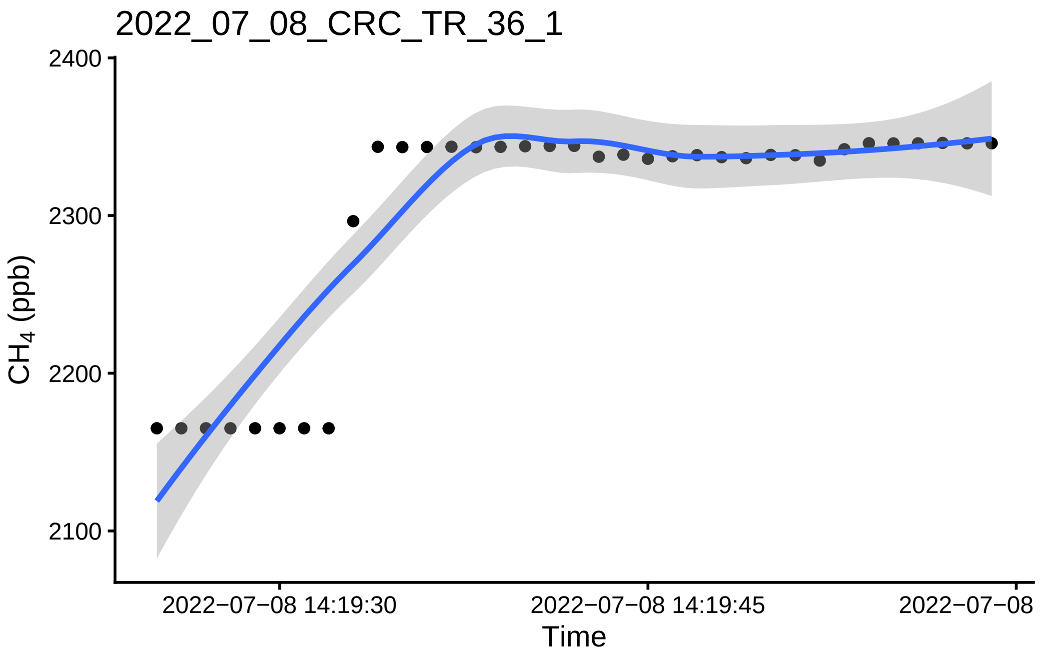
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



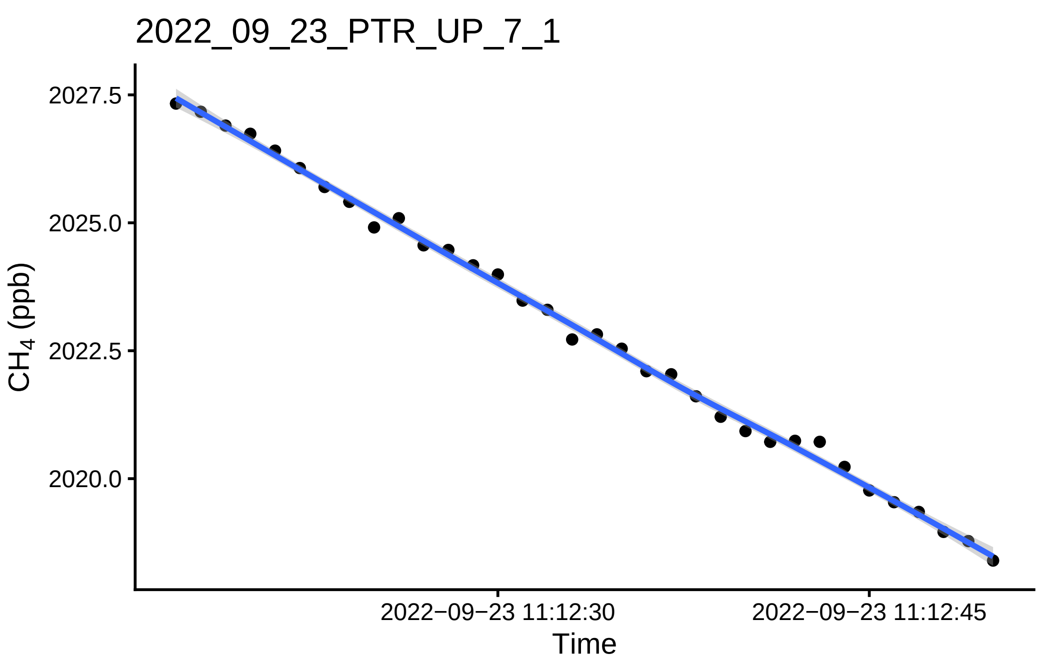
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'



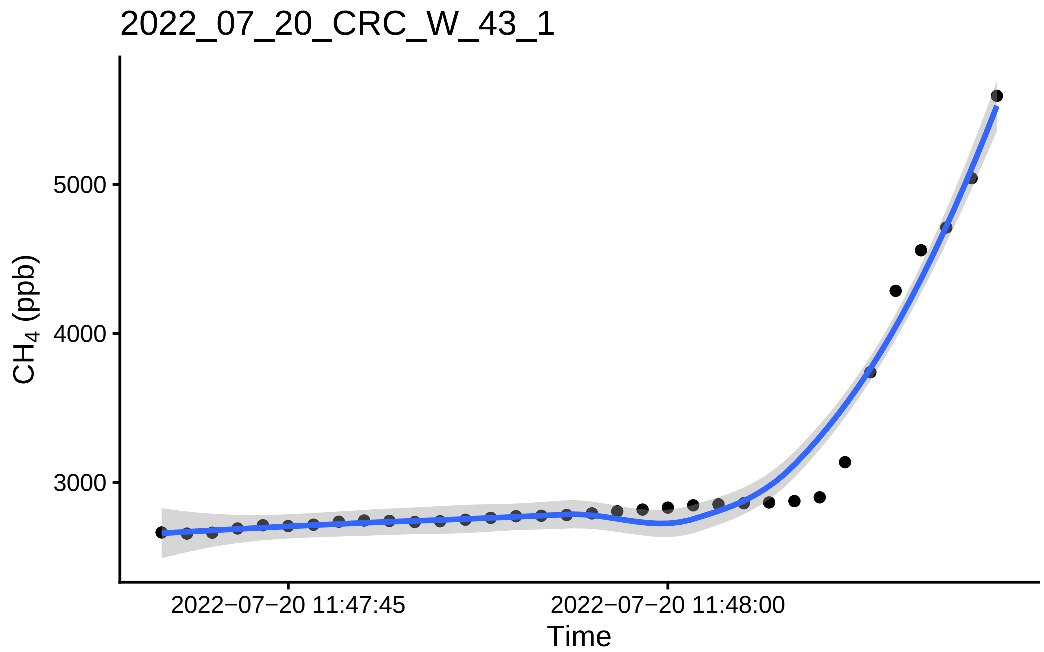
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'




```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

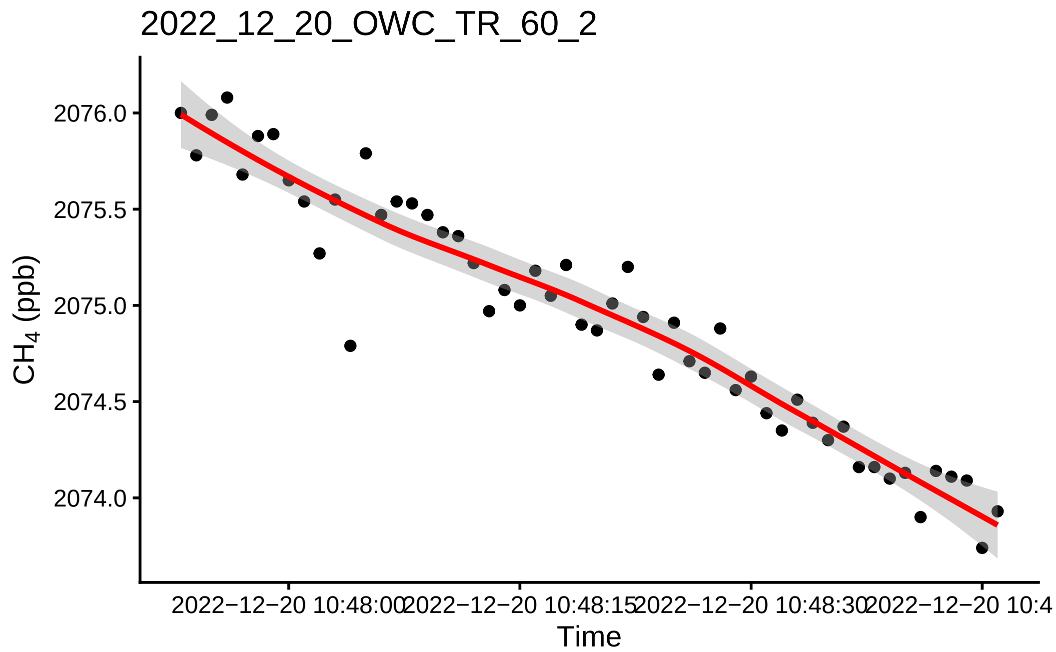



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

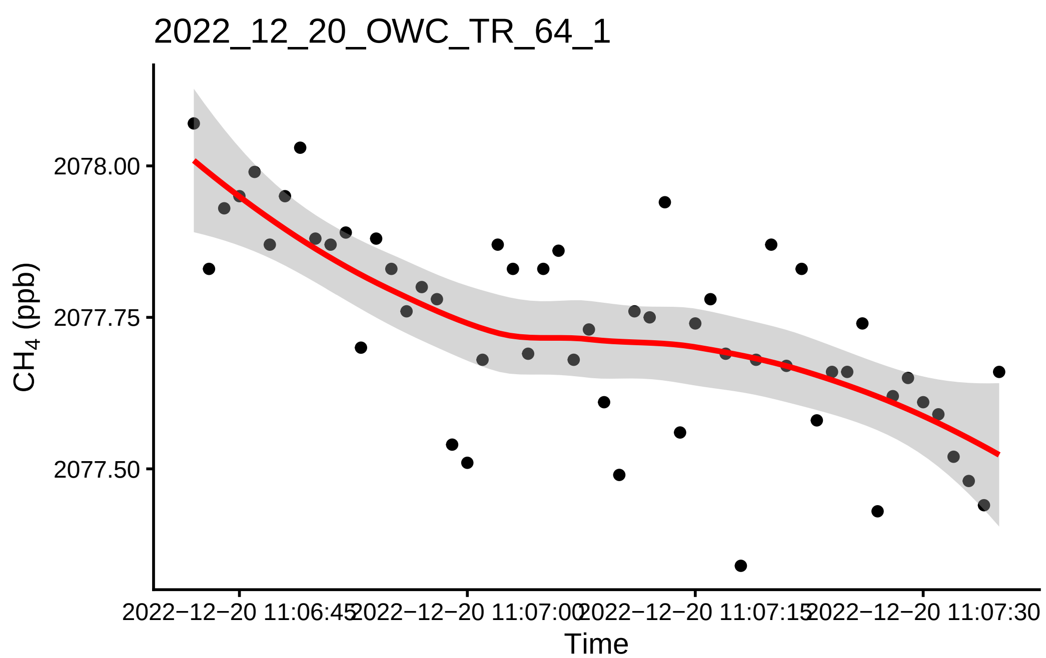


1.2 Check all fluxes

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

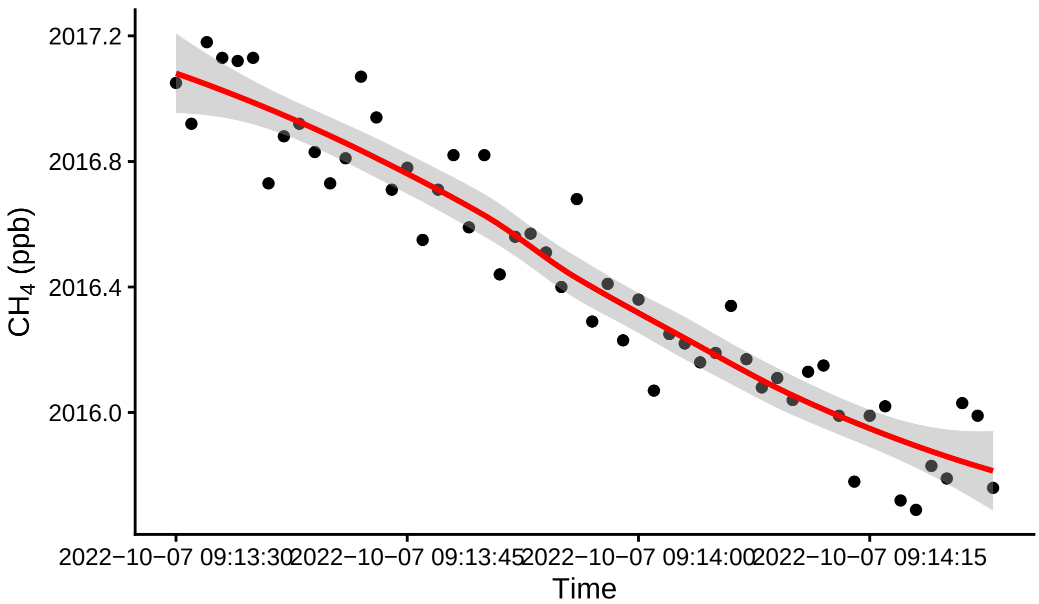



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



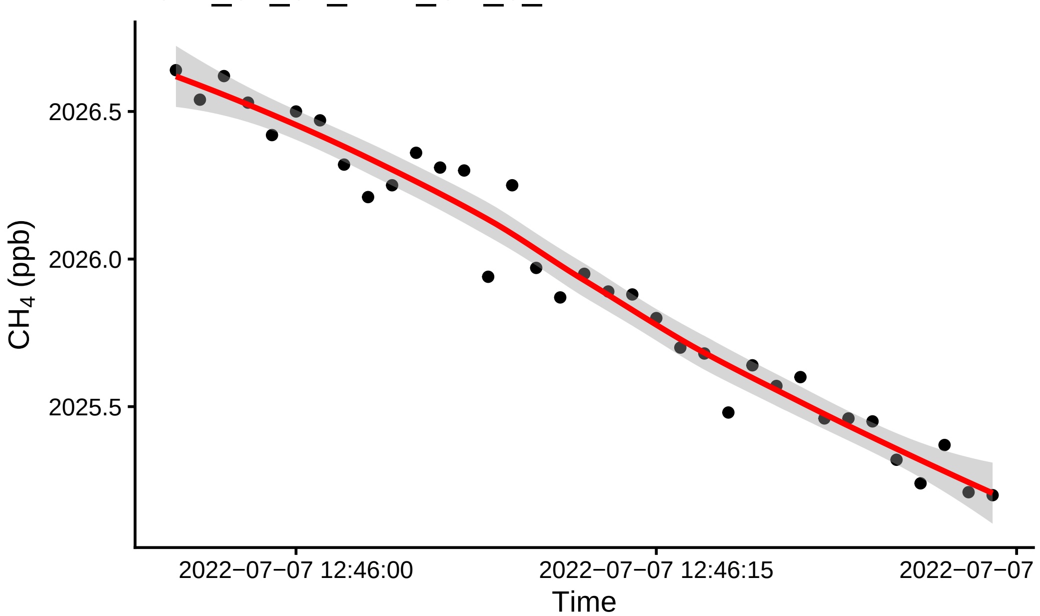

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

2022_10_07_PTR_TR_14_b_2

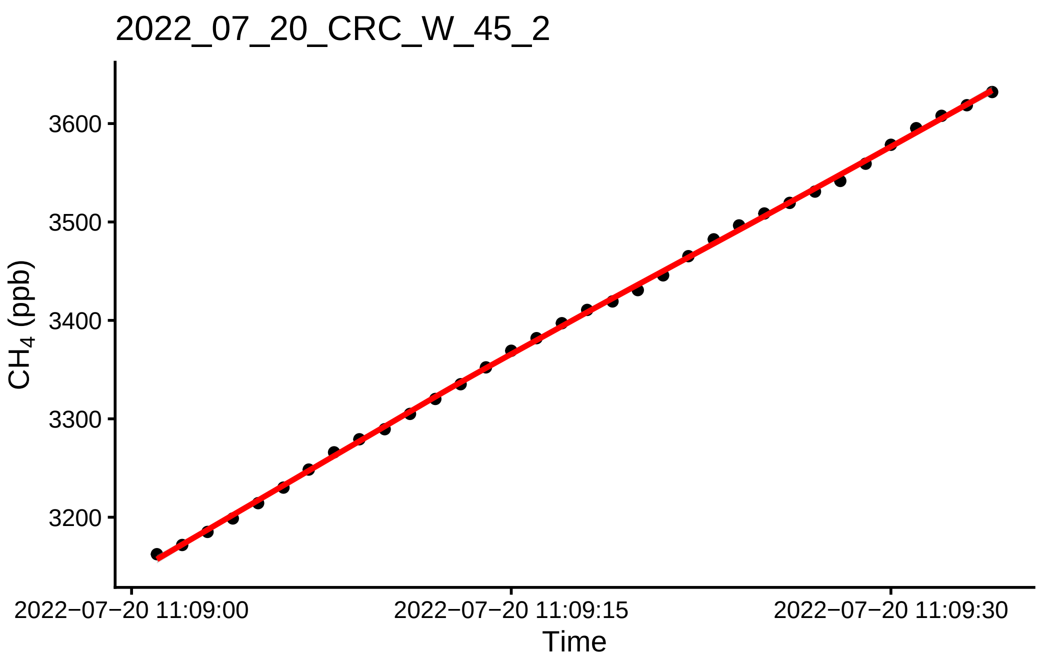



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

2022_07_07_PTR_UP_6_2



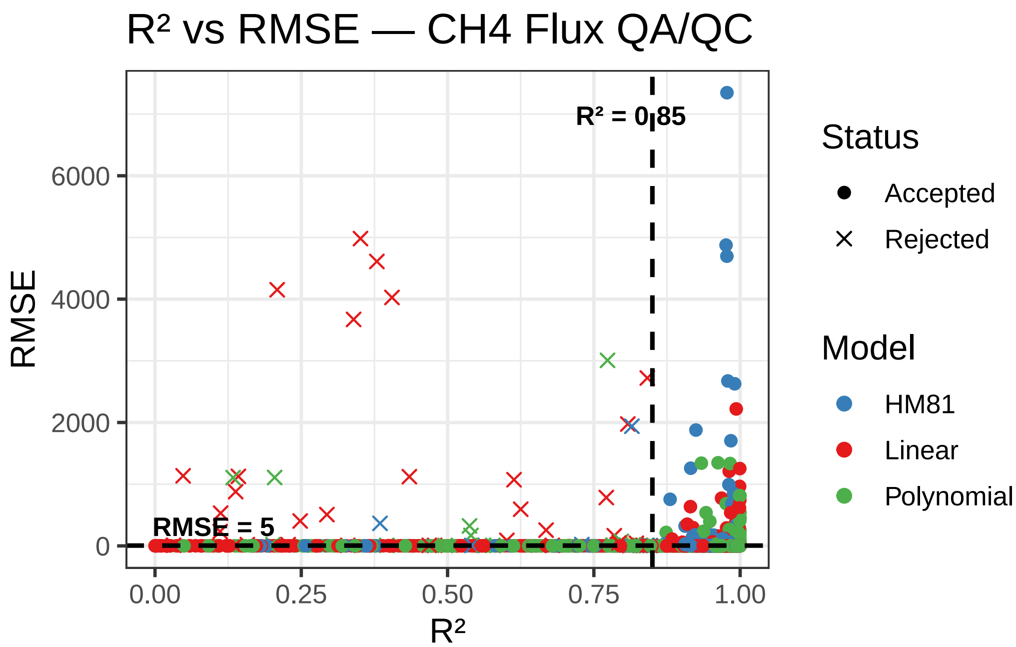

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



1.3 QAQC - Flags

Table 1: QA/QC CH4 Flux Summary

Metric	Value
Total observations	861.0
Rejections	
Rejected — poor model fit (n)	15.0
Rejected — model (%)	1.7
Rejected - [CH4] (n)	31.0
Rejected — [CH4] (%)	3.0
Model performance	
Passed — Linear model (n)	827.0
Passed — HM81 model (n)	396.0
Passed — Polynomial model (n)	846.0

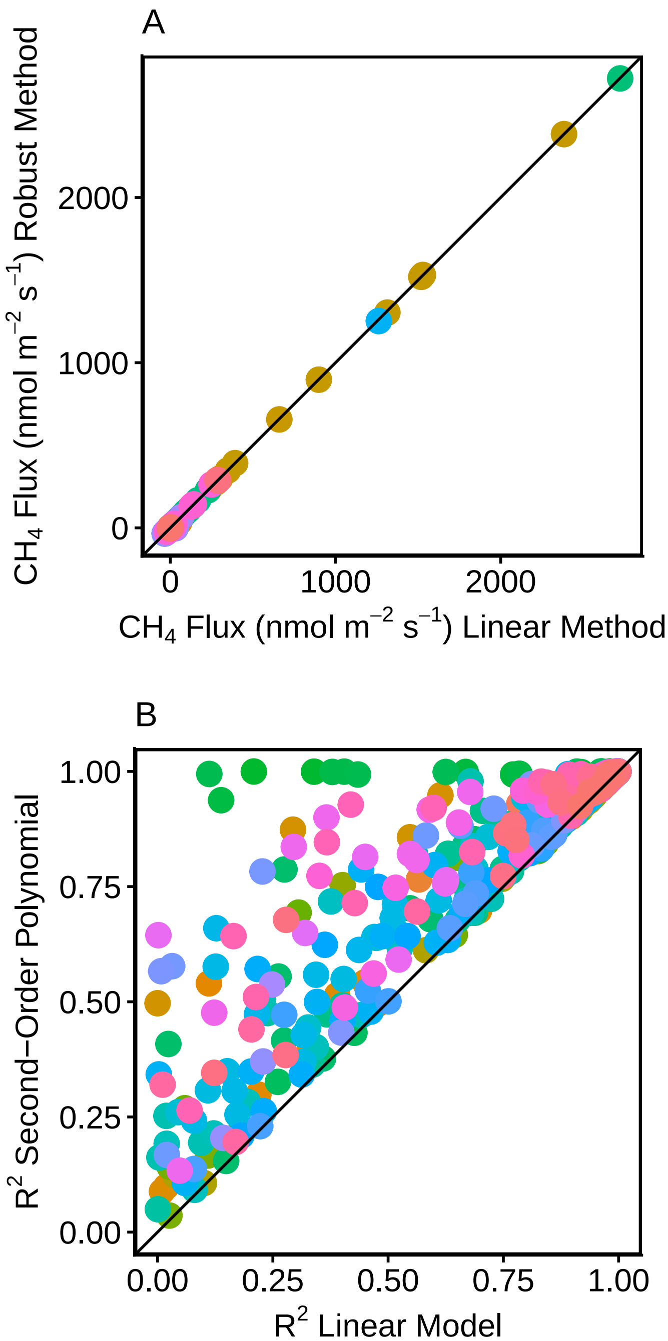


1.4 Adding manual flag to reject any fluxes that do not fit criteria

```
                   Plot_ID           TIMESTAMP HM81_AIC HM81_flux.estimate
1   2022_07_07_PTR_TR_10_1 2022-07-07 13:20:30       NA                 NA
2   2022_07_07_PTR_TR_10_2 2022-07-07 13:23:31       NA                 NA
3   2022_07_07_PTR_TR_11_1 2022-07-07 13:39:57 446.8097          -1.650871
4   2022_07_07_PTR_TR_11_2 2022-07-07 13:42:57 431.5202          -1.174184
5 2022_07_07_PTR_TR_11_b_1 2022-07-07 13:47:41 393.0346          -1.584556
6 2022_07_07_PTR_TR_11_b_2 2022-07-07 13:50:41 395.8692          -1.055917
  HM81_p.value HM81_r.squared HM81_RMSE  lin_AIC lin_flux.estimate
1           NA             NA        NA 208.9624        -0.1477548
2           NA             NA        NA 192.1886        -0.1835150
3 1.887486e-58      0.9785228  5.721301 267.2287        -1.9027854
4 3.754826e-55      0.9731488  5.129381 224.3255        -1.5258896
5 6.226158e-66      0.9870586  3.896537 333.4154        -1.6662026
6 1.439032e-56      0.9756041  3.976233 255.9151        -1.2397587
  lin_flux.std.error lin_int.estimate lin_int.std.error  lin_p.value
1        0.006189500         17938.02         0.3889302 8.960890e-35
2        0.005490624         17642.98         0.3450148 6.222042e-44
3        0.009384338         18390.32         0.5896845 2.456292e-96
4        0.006907381         18311.82         0.4340398 7.292339e-99
5        0.015056424         18108.61         0.9461019 1.645848e-78
```

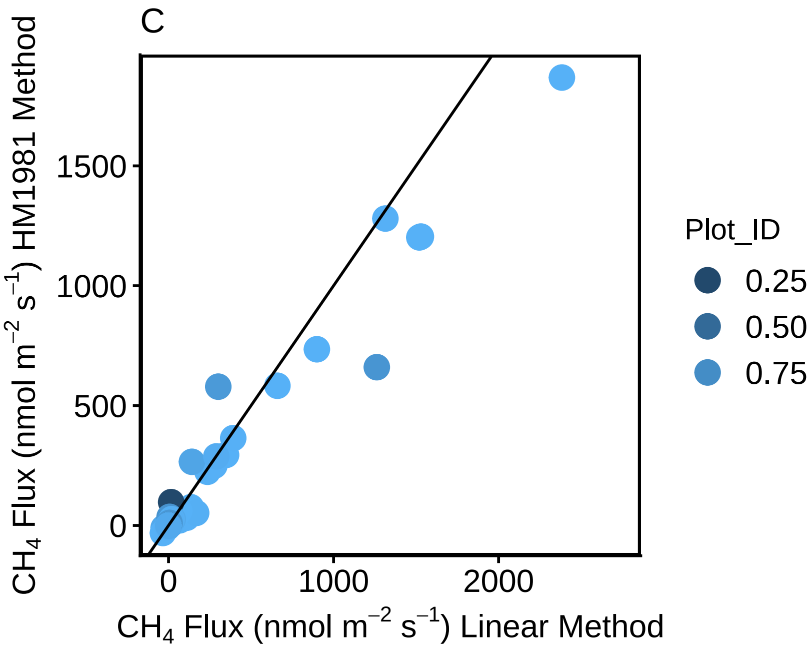
```
6        0.008655808         18182.19         0.5439058 4.277758e-86
  lin_r.squared  lin_RMSE poly_AIC poly_r.squared poly_RMSE  rob_AIC
1     0.8933944 1.0463276 200.7485      0.9104630 0.9733327 209.0010
2     0.9426218 0.9281835 196.0554      0.9427309 0.9412457 192.4418
3     0.9983487 1.5864112 235.4357      0.9990097 1.2469950 267.3751
4     0.9986085 1.1676846 224.3331      0.9986856 1.1519230 224.5513
5     0.9944781 2.5452705 250.3996      0.9984070 1.3876637 333.4375
6     0.9966962 1.4632541 245.1564      0.9973242 1.3366554 255.9602
  rob_converged rob_flux.estimate rob_flux.std.error  rob_RMSE
1          TRUE        -0.1471347        0.006574525 1.2720970
2          TRUE        -0.1858156        0.005495574 0.7695689
3          TRUE        -1.9001307        0.009311501 1.7806284
4          TRUE        -1.5281627        0.006840635 1.3180605
5          TRUE        -1.6641564        0.015616301 3.0599561
6          TRUE        -1.2411887        0.008996664 1.5090392
        TIMESTAMP_min       TIMESTAMP_max    soilp_c   soilp_m soilp_t
1 2022-07-07 13:19:56 2022-07-07 13:21:05 0.01700000 0.2271263    24.7
2 2022-07-07 13:22:57 2022-07-07 13:24:06 0.01725263 0.2267474    24.7
3 2022-07-07 13:39:23 2022-07-07 13:40:32 0.02115789 0.2070000    25.4
4 2022-07-07 13:42:23 2022-07-07 13:43:32 0.02500000 0.2360000    25.6
5 2022-07-07 13:47:07 2022-07-07 13:48:16 0.02400000 0.2380000    25.6
6 2022-07-07 13:50:07 2022-07-07 13:51:16 0.02400000 0.2380000    25.6
  CH4_initial CH4_lin_pass CH4_HM81_pass CH4_poly_pass CH4_init_conc
1     2086.41         TRUE         FALSE          TRUE         FALSE
2     2046.77         TRUE         FALSE          TRUE         FALSE
3     2082.57         TRUE          TRUE          TRUE         FALSE
4     2069.80         TRUE          TRUE          TRUE         FALSE
5     2050.20         TRUE          TRUE          TRUE         FALSE
6     2056.26         TRUE          TRUE          TRUE         FALSE
  CH4_rejected_model CH4_rejected_manual CH4_flux_rejected
1              FALSE               FALSE             FALSE
2              FALSE               FALSE             FALSE
3              FALSE               FALSE             FALSE
4              FALSE               FALSE             FALSE
5              FALSE               FALSE             FALSE
6              FALSE               FALSE             FALSE
```

1.5 Do some visualization of QAQC plots to assess fluxes




```
Warning: Removed 459 rows containing missing values or values outside the scale range
```

(`geom_point()`).



1.6 Read out final dataframe and export for use elsewhere